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PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

ACID ZINC

THE COMPLETE TRAINING MANUAL

A full training course for the brand-new operator — from "what is an anode?" to a signed certificate of completion. Acid chloride zinc (non-cyanide) on steel. Assumes you know nothing. Teaches you everything.

01 CLEAN ▸ 02 RINSE ▸ 03 ACTIVATE ▸ 04 RINSE

05 PLATE ▸ 06 RINSE ▸ 07 PASSIVATE ▸ 08 DRY

OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 · 2026 · FOR-DUMMIES STYLE

FIND YOUR WAY

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READ FRONT-TO-BACK THE FIRST TIME · KEEP ON THE SHELF FOREVER AFTER

★ **REMEMBER**

This manual is a training and reference tool — **not** a substitute for your facility’s written procedures, your supplier’s TDS/SDS, or your supervisor’s instructions. When this manual and your shop’s official procedure disagree, **follow your shop’s official procedure and ask your supervisor.**

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU ARE IN EXACTLY THE RIGHT PLACE

Welcome. If you are holding this manual, you are about to learn how to run — or at least understand — an **acid zinc plating line**. Maybe you started this week. Maybe you have been racking parts for a year and finally want to know *why* you do the things you do. Either way, you are in the right place, and we are glad you are here.

Here is the most important thing we can tell you up front: **this manual assumes you know nothing about plating, chemistry, or electricity.** That is not an insult. That is a promise. Every term gets defined the first time we use it. Every step explains not just *what* to do but *why* it matters. Nobody is born knowing what "cathode efficiency" means or why a rinse tank can quietly ruin a whole shift. You learn it. We are going to teach it.

This manual is written in the spirit of the *For Dummies* books — friendly, plain-spoken, and patient. We would rather over-explain than leave you guessing in front of a tank full of acid.

> ★ **REMEMBER**
> You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: clean it, rinse it, activate it, rinse it, plate it, rinse it, passivate it, and rinse/dry it. Reading it in order matters, because (as you'll soon learn) **the order of the plating line is not random** — and neither is the order of this book.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

As you read, you will run into little flagged notes. Here is what each one means.

> 💡 **TIP**
> A shortcut, a trick of the trade, or a smarter way to do something — the kind of thing a 20-year veteran would lean over and tell you. These save you time and headaches.

> ★ **REMEMBER**
> A core idea worth locking in. If you forget everything else on the page, don't forget these. They're the load-bearing ideas.

> ⚠ **HEADS-UP**
> A safety warning. Read every single one. These are the notes that keep your skin, your eyes, and your lungs intact. We use them *liberally* because an acid zinc line has real hazards.

> ⚙ **TECHNICAL STUFF**
> Deeper chemistry or theory for the curious. Totally optional. Skip these and you can still run the line perfectly. Read them and you will understand it on a deeper level.

> ☀ **FUN FACT**
> A bit of interesting trivia to make the science stick (and to give you something to say at lunch).

> 💡 **TIP**
> Read the Fundamentals all the way through *before* you read any individual stage chapter. The stage chapters assume you already understand the big picture this section gives you. It's the difference between reading a recipe when you already know how to cook versus the very first time you've stepped into a kitchen.

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Besides the margin icons, each station chapter ends with two recurring tools. The first is a **Teach-Back** checklist (sometimes paired with a red "Top Mistakes" card) — the things you must be able to *say out loud, unprompted* before you're cleared on that station; it's how you and your trainer confirm the ideas actually stuck. The second is a bordered **Sign-Off** box at the very end of the station — the short statement your *trainer initials* (with you) once you've demonstrated the skill safely. Treat the Teach-Back as your study guide and the Sign-Off as the gate you have to pass through.

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

THE WHOLE FOUNDATION — MASTER THIS AND EVERY STAGE CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what electroplating is** in plain language — how electricity moves metal from one place onto another.
2. **Describe what acid zinc plating does** and why a shop chooses it — bright finish, fast deposition, and sacrificial corrosion protection on steel.
3. **Name all eight stages of the line in order** and explain *why* the order cannot be rearranged.
4. **Understand what each tank is for** — what goes in, what comes out, and what "good" looks like at every handoff.
5. **Recognize the main hazards** at each station — acids, alkali, electrical, mist, and chromate chemistry — and know the correct PPE and emergency response for each.
6. **Use the basic field tests** every operator relies on, like the water-break test and conductivity checks.
7. **Spot trouble early** — read a deposit and know whether the problem is upstream, in the bath, or in your handling.
8. **Talk the language** — use the right words so you can communicate clearly with your lead, your chemist, and your chemistry supplier.

> ★ **REMEMBER**
> You are not expected to hit all of these on day one. Plating is a craft. The goal is steady competence, not instant mastery.

• THE EIGHT-STAGE PROCESS AT A GLANCE

Here is the whole acid zinc line in one strip. Don't memorize it yet — just get a feel for the rhythm: **Clean → Rinse → Activate → Rinse → Plate → Rinse → Passivate → Rinse/Dry.**

> ★ **REMEMBER**
> Notice the pattern: **a rinse follows almost every working tank.** Rinses are not breaks — they are checkpoints that keep one chemistry from poisoning the next. Every working chemistry wants to "hitch a ride" into the next tank on the surface of your part, and rinses are what stop that. They aren't filler between the "real" steps; they're some of the most important steps you'll perform.

CHAPTER 1

WHAT IS ELECTROPLATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT PLATING IS

THE ONE-SENTENCE VERSION

Electroplating is using electricity to coat one metal with a thin layer of another metal. That's the whole idea. Now let's unpack it so it actually means something.

Imagine you have a steel bolt. Steel is strong and cheap, but it rusts. You want to give that bolt a thin protective skin of **zinc** — a metal that protects steel from rust (we'll explain exactly how in a minute). You *could* try to paint zinc on, but that would be uneven and weak. Instead, we use electricity to grow a perfectly even, tightly-bonded layer of zinc directly onto the steel, atom by atom. That process is electroplating.

HOW ELECTRICITY MOVES METAL: THE KITCHEN-SINK PICTURE

Here is how it physically works. Picture a tank full of liquid called the **electrolyte** (or just "the bath"). This liquid has zinc dissolved in it — invisible zinc, broken down into tiny charged particles called **ions**. An *ion* is just an atom that carries an electrical charge. Dissolved zinc floats around the bath as positively-charged zinc ions, written **Zn^{2+}**.

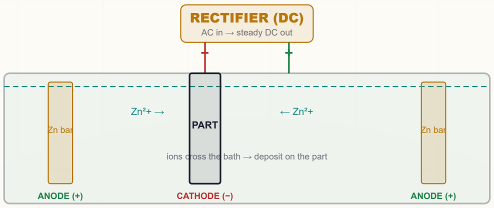
You hook two things up to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction "DC" current plating needs):

- **The cathode** — this is your *part* (your steel bolt), connected to the **negative** terminal. Just remember "cathode = the part, the thing collecting metal."
- **The anode** — bars of fresh zinc hanging in the same tank, connected to the **positive** terminal. As the bath uses up its dissolved zinc, the anodes slowly dissolve to replace it — they are the "fuel tank" of zinc metal. ("Anode = positive = the metal that feeds the bath.")

When you turn on the rectifier, here is the dance that happens:

1. Current flows. The positively-charged zinc ions (Zn^{2+}) in the bath are *attracted* to the negatively-charged part (the cathode), because opposite charges attract.
2. When a zinc ion reaches the part's surface, it grabs electrons from the part and turns back into solid zinc metal, sticking to the surface.
3. Meanwhile, at the anodes, solid zinc dissolves *into* the bath, releasing new Zn^{2+} ions to keep the supply topped up.

The net result: zinc moves off the anodes, through the bath, and onto your part. The longer you plate and the more current you push, the thicker the coating.



REMEMBER

Three words to lock in now: **Cathode = the part** (collects zinc). **Anode = the zinc bars** (feed the bath). **Electrolyte = the bath** (carries zinc from one to the other). Almost everything in plating is a variation on this one idea.

TECHNICAL STUFF

At the part, a **reduction**: $Zn^{2+} + 2e^- \rightarrow Zn^0$. At the anode, the reverse — solid zinc gives up two electrons and dissolves: $Zn^0 \rightarrow Zn^{2+} + 2e^-$. Because the anodes replenish what the cathode consumes, a well-run acid zinc bath is largely self-balancing on zinc metal.

FUN FACT

"Galvanize" and the unit "volt" trace back to Luigi Galvani and Alessandro Volta, two 1790s Italian scientists who argued over whether electricity came from frog legs or from metals. Volta won — and you'll read volts on your rectifier every day.

HEADS-UP

Plating rectifiers push enormous current — hundreds to thousands of amps — low voltage but very high current, into a tank full of conductive liquid. **Never reach into an energized tank, and never do maintenance until equipment is locked and tagged out (LOTO).** Electricity plus salt water plus your hand demands total respect.

CHAPTER 2

WHAT IS “ACID ZINC”?

AND WHY DO WE USE IT — FAST, BRIGHT, SACRIFICIAL PROTECTION ON STEEL

ZINC PLATING COMES IN FAMILIES

“Zinc plating” just means coating something with zinc. But there are *several different recipes* for the bath that does it. The two big families are:

- **Alkaline zinc** — a bath that is chemically *basic* (high pH, soapy-feeling, the opposite of acidic). Often cyanide-based in older shops, or cyanide-free in modern ones.
- **Acid zinc** — a bath that is chemically *acidic* (low pH). This manual is about acid zinc.

TECHNICAL STUFF

pH is the scale we use to measure how acidic or basic a liquid is. It runs from 0 (strong acid) to 14 (strong base), with 7 being neutral (pure water). Lower number = more acidic. Our acid zinc plating bath runs at **pH 4.8–5.8** — mildly acidic, roughly like black coffee or a tomato. Not a screaming-strong acid, but acidic enough to matter.

More precisely, our process is **acid chloride zinc**. “Chloride” tells you the main supporting chemistry: the bath is loaded with a chloride salt — either **potassium chloride (KCl)** or **ammonium chloride (NH_4Cl)** — dissolved alongside the zinc. That chloride does several jobs: it carries electrical current efficiently, it helps the anodes dissolve evenly, and it shapes how the zinc deposits.

THE BATH AT A GLANCE

COMPONENT	TYPICAL RANGE	WHAT IT DOES
Zinc metal (as $ZnCl_2$)	15–45 g/L (2–6 oz/gal)	The metal you’re plating
KCl *or* NH_4Cl	120–250 g/L (16–33 oz/gal)	Carries current; helps anodes dissolve
Boric acid (H_3BO_3)	20–35 g/L (2.7–4.7 oz/gal)	A pH “buffer” — keeps acidity steady at the part’s surface
Carrier (conditioner)	Per supplier’s TDS	Refines grain, wets the surface, prevents pitting
Primary brightener	Per supplier’s TDS	Makes the deposit shiny
Secondary brightener	Per supplier’s TDS	Adds shine and leveling in high-current areas
pH	4.8–5.8	Adjusted with HCl (to lower) or KOH (to raise)

TECHNICAL STUFF

“g/L” means **grams per liter** — how many grams of a substance are dissolved in each liter of bath. “oz/gal” is the same idea in ounces per gallon, which many US shops still use. A “**TDS**” is a **Technical Data Sheet** — the instruction document your chemistry supplier provides for each product. When this manual says “per TDS,” it means *follow the supplier’s sheet*, because brightener and additive amounts are specific to each product line. Never guess at additive levels.

REMEMBER

Boric acid is not optional. It looks like a minor ingredient, but it is the buffer that holds the right acidity in the thin film of liquid right against the part where plating actually happens. Neglect the boric acid and your deposit quality falls apart even when everything else looks fine.

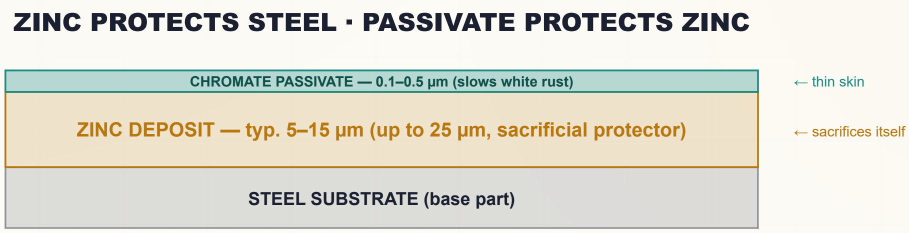
• WHY DO SHOPS USE ACID ZINC?

Two big reasons: **corrosion protection** and **appearance** — delivered **fast** and **efficiently**. Let's take them one at a time.

REASON 1: SACRIFICIAL CORROSION PROTECTION

Steel rusts. Rust (iron oxide) is what happens when iron in the steel reacts with oxygen and moisture. Rust is weak, flaky, and spreads — it destroys parts. Zinc protects steel in a clever way. It is not just a physical barrier (though it is that too). Zinc is what we call a **sacrificial coating**. Zinc is *more chemically reactive* than steel — it "wants" to corrode more eagerly than iron does. So when a zinc-coated steel part gets wet and starts to corrode, the **zinc corrodes first and protects the steel underneath.** Even if the coating gets scratched and bare steel is exposed, the surrounding zinc keeps protecting that scratched spot, because the zinc corrodes preferentially.

THE FINISH, LAYER BY LAYER (NOT TO SCALE)



TECHNICAL STUFF

This is called **galvanic** or **cathodic protection**. In an electrochemical pairing, the more *active* (more easily oxidized) metal becomes the **anode** and corrodes, while the less active metal becomes the **cathode** and is protected. Zinc is more active than iron, so zinc takes the hit. This is why a scratch in zinc plating doesn't immediately rust the way a scratch in paint or in nickel plating would — the zinc keeps "throwing itself on the grenade" for the exposed steel nearby.

FUN FACT

Same principle as the chunky "sacrificial anodes" bolted to boat hulls and water heaters — blocks of zinc or magnesium that corrode away so the steel doesn't. Your plated bolt carries a microscopic version of the same bodyguard.

REMEMBER

Zinc protects steel by sacrificing itself. That single idea is the entire reason this line exists. Everything you do on this line — every clean, rinse, and dip — is in service of laying down a good, even layer of that sacrificial zinc.

HOW THICK SHOULD THE ZINC BE? (SERVICE CONDITIONS)

How much zinc a part needs depends on how harsh a life it will lead. The industry standard **ASTM B633** sorts parts into four **Service Conditions (SC)** — from a screw living indoors in a dry drawer to a bracket fighting salt spray on a truck frame. The harsher the environment, the thicker the minimum zinc. Typical commercial acid zinc runs **5–15 µm**; severe-service parts go up to **25 µm**.

SERVICE CONDITION	ENVIRONMENT	MIN. ZINC THICKNESS
SC 1 — Mild	Indoor, dry (warm rooms, sheltered)	5 µm
SC 2 — Moderate	Indoor, some condensation/humidity	8 µm
SC 3 — Severe	Outdoor, moderate weather exposure	12 µm
SC 4 — Very Severe	Outdoor harsh / marine / road salt	25 µm

TIP

You don't pick the Service Condition — the customer's spec or print does. Your job is to plate long enough to hit the minimum thickness for the SC on the job ticket. When in doubt, ask your lead which SC the part is being run to before you start the clock.

REASON 2: A BRIGHT, ATTRACTIVE FINISH

Acid zinc, with the right brightener system, comes out of the bath **mirror-bright** — shiny straight from the tank, no polishing required. The chloride chemistry produces a very fine-grained, smooth, leveled deposit. "Leveling" means the deposit fills in tiny scratches and low spots, making even a so-so surface look better. That's why acid zinc is popular for parts people actually *see* — decorative hardware, fasteners, brackets — not just hidden structural pieces.

REASON 3: SPEED AND EFFICIENCY

This is where acid zinc really shines compared to alkaline zinc. The key number is **cathode efficiency: typically about 95–98%** (supplier- and condition-dependent — some cite up to ~99%; it drops at low current density and as the bath drifts).

Cathode efficiency means: of all the electrical current you push into the part, what percentage actually goes into depositing zinc (versus being wasted)? In a healthy acid zinc bath, roughly 95–98% of your current makes zinc — almost none is wasted — though the exact figure depends on the chemistry, temperature, and current density. By comparison, alkaline zinc baths typically run only about **60–80%** efficient — they waste a lot of current splitting water into hydrogen gas instead of plating zinc.

Higher efficiency means **acid zinc plates 2–3 times faster** than alkaline zinc at the same current, and uses less electricity per part. More parts per shift, lower energy bill. It also typically runs at or near **room temperature (70–95°F)**, so you're not paying to heat a big tank.

> ★ **REMEMBER**
> The headline number for this whole process is **95–98% cathode efficiency.** Nearly every amp-hour of electricity becomes zinc on the part. As the saying goes: *"95% efficiency means 95% of your current is making money. Control the other 5%."*

> ⚠ **HEADS-UP**
> That same high efficiency has a hidden cost you must respect: because so little current is "wasted" making hydrogen gas, more atomic hydrogen gets absorbed *into the steel* per unit of zinc deposited. On hard, high-strength steel this can cause **hydrogen embrittlement** — a dangerous weakening of the part. We cover this in detail in the safety section and the relevant stage chapters. For now, just file away: *acid zinc is fast, and fast has a price on hard parts.*

THE HONEST TRADEOFFS

A good operator knows the weaknesses too. Acid zinc is not the answer for everything:

- **Poor throwing power.** "Throwing power" is the bath's ability to plate evenly into deep recesses, blind holes, and the insides of tubes. **Acid zinc covers recesses noticeably worse than alkaline zinc** — parts with deep cavities may come out thin or bare inside. (The shop numbers you'll hear — roughly 3:1 to 5:1 for acid vs. 1.5:1 to 2:1 for alkaline — are surface-to-recess thickness ratios, so a number *closer to 1:1 is better, more even coverage*. Acid zinc's bigger spread is the worse one.)
- **Very sensitive to metallic contamination.** Tiny amounts of iron, copper, or lead getting into the bath cause dark deposits, spots, and roughness. Acid zinc demands tighter control than alkaline.
- **Complex, breakdown-prone brighteners.** The three-part brightener system (carrier + primary + secondary) needs attention.
- **Corrosive to equipment.** Chloride attacks bare steel — tanks, heaters, and bus bars must be plastic-lined or plastic, never bare mild steel.

> 💡 **TIP**
> If you're ever told "this part keeps coming out thin in the recesses," your first thought should be *throwing power* — and whether acid zinc was even the right process for that geometry. That's not your fault as an operator; it's a process-selection reality.

A WORD ABOUT THE FINAL STATION: PASSIVATION

The acid zinc line ends with **trivalent chromate passivation**. After plating, you have bright bare zinc. Left alone, that zinc oxidizes in humid air and forms a powdery white corrosion product called **white rust**. To prevent it, we dip the part in a **chromate conversion coating** — a chemical bath that reacts with the zinc surface to grow a thin, tough, protective film. This film slows white rust dramatically and can give the part a color: **clear/blue**, **yellow**, or **black**, depending on the product. Modern passivates use **trivalent chromium**, written **Cr(III)** or **Cr^{3+}** — the RoHS-compliant, far less hazardous form.

> ⚠ **HEADS-UP**
> The old generation of passivates used **hexavalent chromium — Cr(VI), or Cr^{6+}** — which is a confirmed **human carcinogen** and is heavily restricted. Most modern shops have switched to trivalent. **Know which one your shop uses.** If any hexavalent chromate is still in use, it carries strict OSHA requirements (PEL of 5 µg/m³, supplied-air respirators, medical surveillance per 29 CFR 1910.1026) that go far beyond ordinary line PPE.

> ★ **REMEMBER**
> **Zinc without passivate is a coating waiting to white rust.** Passivation isn't an optional finishing touch — it's what makes the zinc layer actually last in service.

CHAPTER 3

HOW A PLATING LINE WORKS

THE BIG PICTURE — A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

A LINE IS A SEQUENCE OF TANKS

An acid zinc line is not one tank. It's a **sequence of tanks** that a part travels through in a strict order. Think of it like a kitchen line or a car wash: each station does one job, and each job prepares the surface for the next. Parts ride down the line in one of two ways:

- **Rack plating** — Parts are individually hung on a **rack** (a frame with hooks or clips that holds parts and carries electrical current to them). Used for larger or delicate parts.
- **Barrel plating** — Lots of small parts (screws, washers, small fittings) are dumped together into a perforated rotating **barrel** that tumbles them through the chemistry. Far more efficient for high volumes of tiny parts.

💡 **TIP**

You'll see different numbers for the same chemistry depending on rack vs. barrel — barrel baths often run a bit higher in zinc and chloride to make up for the lower current density tumbling parts receive. When in doubt about which numbers apply, check whether you're running rack or barrel and read the matching column on your shop-floor poster.

• THE WHOLE ACID ZINC SEQUENCE: 8 STAGES

Here is the complete process. Don't memorize the details — just get a feel for the *flow.* Each stage gets its own full chapter later.

#	STAGE	THE ONE THING IT DOES	TEMP	TIME
01	Alkaline Soak Clean	Removes all oil, grease, and shop soil	130–160°F	3–10 min
02	Rinse (Pre-Activation)	Washes off cleaner before the acid dip	Ambient	30–60 s
03	Acid Activation / Pickle	Dissolves surface oxide; "wakes up" the steel	Amb–110°F	15–60 s
04	Rinse (Pre-Plate)	Guards the plating bath from acid & iron drag-in	Ambient	30–60 s
05	Acid Zinc Plating	Deposits the bright zinc layer	70–95°F	5–40 min
06	Rinse (Post-Plate)	Washes off chloride before passivation	Ambient	30–60 s
07	Chromate / Passivate	Adds a chromate skin for extra corrosion protection	70–110°F	30–90 s
08	Final Rinse / Dry	Cleans up and dries without damaging the coating	Amb–150°F	30–60 s + dry

The basic rhythm is easy to remember: **Clean** → **Rinse** → **Activate** → **Rinse** → **Plate** → **Rinse** → **Passivate** → **Rinse/Dry.**

• A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

★ **REMEMBER**

Every stage either protects the next stage or poisons it. There is no neutral step. A part that leaves the cleaner still carrying a film of cleaner will contaminate the acid. A part that leaves the acid still wet with acid will contaminate the zinc bath. What you do (or fail to do) at Stage 1 shows up as a defect at Stage 5. The line is a chain, and a chain is only as strong as its weakest link.

This is also why **the plating bath itself is so precious.** Unlike the cleaner and the acid (dumped and remade regularly), a well-maintained acid zinc bath can run a very long time. But here's the catch unique to acid zinc: contaminants such as dissolved **iron** that sneak in **stay in solution and accumulate** — the bath can't simply filter them out the way an alkaline bath can. Every milligram of junk you let drag into that tank is a permanent guest.

💡 **TIP**

Think of the acid zinc tank as the crown jewel of the line. Everything *before* it (clean, rinse, activate, rinse) exists largely to ensure only a perfectly clean, bare, well-rinsed part ever touches it. Everything *after* it (rinse, passivate, rinse/dry) exists to protect and finish what the zinc created. The whole line orbits that one tank.

CHAPTER 4

WHY THE ORDER MATTERS

WALKING THE LINE — EACH STAGE SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY

You might think the stage order is just convention — that you could shuffle steps or skip one to save time. You can't. Each stage exists *because of* the stage before and after it. Here is the logic, link by link.

STAGE 1 — CLEAN: GET RID OF THE JUNK FIRST

Plating, acid, and chromate all need to touch *bare metal*. Oil and shop soil block them. **Plate over dirt and you've just plated dirt** — the zinc will bond to the grease, not the steel, and peel off later. There is a free, foolproof test for cleanliness called the **water-break test** (see the cleaning chapter): if water sheets off in an unbroken film, the part is clean; if it beads up like on a waxed car, oil remains.

> 💡 **TIP**
>
> The simplest, most powerful quality check in the whole shop is the **water-break test**, and it's free. Pull a cleaned part and watch how water sheets off it. A smooth, continuous, unbroken sheet ("zero seconds" to break) means the surface is clean. If water beads up or "breaks" into droplets, there's still oil there — *stop and re-clean.*

> ⚠ **HEADS-UP**
>
> The cleaner is **hot (130–160°F) and strongly alkaline (pH 11–13.5)** — it causes chemical burns and can seriously injure your eyes. Wear full PPE, and know where your shower and eyewash are before you need them.

STAGE 2 — RINSE (PRE-ACTIVATION): DON'T CARRY THE CLEANER FORWARD

The cleaner is alkaline (basic). The next tank is acid. If you carry cleaner straight into the acid, it neutralizes the acid, wastes it, and weakens the activation. The rinse strips the cleaner film off first.

> ★ **REMEMBER**
>
> *"Drag cleaner into acid and you waste acid. Drag cleaner into the zinc bath and you poison it."* Rinses exist to keep each chemistry from invading the next tank. The cleaner-to-acid rinse is your first firewall on the line.

STAGE 3 — ACID ACTIVATION: WAKE UP THE METAL

Even a clean steel surface grows an invisible layer of **oxide** (a tarnish-like film from reacting with air). Zinc can't bond to oxide. The acid dip dissolves that oxide and leaves a chemically "fresh," active steel surface for the zinc to grip.

- **Under-dip** (too short/weak) = oxide remains = zinc peels off later (adhesion failure).
- **Over-dip** (too long/strong/hot) = you start etching and pitting the part itself, and load it with hydrogen.

★ **REMEMBER**

No activation, no adhesion. Period. Activation has to come right before plating, with only a rinse in between — because freshly activated bare metal just grows a new oxide skin if it sits.

⚠ **HEADS-UP**

The acid dip uses **strong mineral acid** — typically HCl at 10–50% or H_2SO_4 at 5–25%. These cause severe burns and eye damage, and HCl gives off corrosive, choking fumes even at room temperature. The acid eating into metal also releases **flammable hydrogen gas.** Full acid PPE, good ventilation, and **always add acid to water, never water to acid.**

⚙ **TECHNICAL STUFF**

Hydrogen embrittlement is a real, serious risk. During pickling and plating, atomic hydrogen can diffuse into steel's crystal structure. In high-strength steel (above ~40 HRC, or 180 ksi tensile), trapped hydrogen can cause catastrophic cracking later, under load — sometimes hours or days after the part is in service. The fix is **baking**: 375 ±25°F for at least 8 hours, started **within 4 hours** of the last plating step (ASTM B850). For prone parts, use *inhibited* acids and minimum dip time. A failed bolt or structural part can hurt people — always check the customer spec.

STAGE 4 — RINSE (PRE-PLATE): THE MOST IMPORTANT RINSE ON THE LINE

This is the **guard rinse** protecting your most valuable tank. Acid dragged into the plating bath lowers its pH, breaks down expensive brighteners, and — critically — carries dissolved **iron** from the pickle straight into the zinc bath. Unlike alkaline zinc, an acid zinc bath *can't* filter that iron out; it stays dissolved and accumulates. This rinse is your filter, so it targets a tighter conductivity (**< 200 µS/cm**) and double rinses are strongly recommended.

★ **REMEMBER**

"Every contaminant you allow through this rinse becomes a permanent guest in the zinc bath." The pre-plate rinse is the single most important contamination barrier on the entire line. Drain well, rinse hard, and move the part quickly to the zinc tank before fresh oxide can form.

⚙ **TECHNICAL STUFF**

The recurring villain that justifies all those rinses is **drag-out** (and its mirror image, **drag-in**). Drag-out is the film of solution that clings to a part as it leaves a tank; when carried into the next tank it becomes drag-in — contamination. Operators measure rinse cleanliness with **conductivity**, in microsiemens per centimeter (µS/cm) — basically how many dissolved ions (contamination) are in the water. *Higher conductivity = more dragged-in chemistry = a dirtier rinse.* Critical rinses on this line target **under 200 µS/cm.**

STAGE 5 — ACID ZINC PLATING: THE MAIN EVENT

Now — and only now, with a perfectly clean, freshly activated, well-rinsed part — does the actual zinc plating happen. This is the payoff stage where the bright zinc goes on. The bath is acidic chloride chemistry (pH 4.8–5.8) running near room temperature (70–95°F), with the dissolving zinc anodes feeding fresh metal back into solution as you plate.

> ⚠ **HEADS-UP**
>
> The plating bath is acidic and full of **chloride**, and air agitation throws up a fine **zinc chloride mist** you must not breathe — local exhaust ventilation is required. The rectifier runs **high amperage** — shock and arc-flash risk. Because acid zinc runs at high efficiency, more hydrogen soaks into hard steel per amp-hour, raising embrittlement risk on parts above 40 HRC. LOTO before any rectifier work.

STAGE 6 — RINSE (POST-PLATE): PROTECT THE PASSIVATE

The plating bath is full of **chloride**. Drag chloride into the passivate tank and it poisons the chromate chemistry, shortening bath life and producing thin, blotchy coatings.

> ★ **REMEMBER**
>
> **Chloride into chromate is a slow death for the passivate bath.** This rinse is the wall between the two. It's also the decision point for the hydrogen-embrittlement bake on hardened steel.

STAGE 7 — PASSIVATE: LOCK IN THE FINISH

Bare fresh zinc will form ugly **white rust** quickly on its own. The passivate (a chromate conversion coating) adds a thin protective skin that dramatically slows white rust and can add color — clear/blue, yellow, or black. Modern lines use trivalent chromium, Cr(III).

> ⚠ **HEADS-UP**
>
> Even trivalent passivate is **acidic (pH ~1.5–3.5)** and causes burns. If any **hexavalent chromium (Cr^{6+})** is in use, it is a **confirmed human carcinogen** (OSHA PEL 5 μg/m³) requiring supplied-air respirators and medical surveillance under OSHA 29 CFR 1910.1026. Know which chemistry your shop runs.

STAGE 8 — FINAL RINSE / DRY: SEND IT OUT RIGHT

A last clean rinse removes passivate residue, and a *gentle* dry (not too hot) finishes the part without cracking the fresh chromate film. Trivalent films crack above 150°F, so don't over-dry.

> ★ **REMEMBER**
>
> The whole eight-stage sequence is one continuous logic chain: *clean it so zinc can stick → rinse so the cleaner doesn't ruin the acid → activate so the metal is bare and ready → rinse so the acid and iron don't poison the precious zinc bath → plate the bright zinc → rinse so chloride doesn't kill the passivate → passivate to protect the zinc → final rinse and gentle dry so it ships flawless.* Every step earns its place. Skip one, and the chemistry of the next step tells on you.

> 💡 **TIP**
>
> When you understand *why* each stage exists, troubleshooting becomes logical instead of mysterious. A peeling deposit? Look upstream at cleaning and activation. A dying passivate bath? Look at the post-plate rinse. The defect almost always points back to the stage that was supposed to prepare for it.

CHAPTER 5

PPE — YOUR DAILY ARMOR

THE GEAR THAT STANDS BETWEEN YOU AND CHEMICALS THAT BURN, MIST THAT HARMS, SPLASHES THAT BLIND

Now the part that keeps you safe. An acid zinc line works with hot caustic, strong mineral acids, electrical current at high amperage, chemical mists, and chromium chemistry. None of these will hurt you if you respect them and wear the right gear. All of them can hurt you badly if you don't.

★ **REMEMBER**

There is no part, no order, and no deadline worth an injury. Scrap can be reworked. Your eyes cannot.

• THE MASTER PPE SET (WEAR AT ALL STATIONS)

Treat this as your baseline "uniform" any time you are on the line:

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; a splash finds the gaps.
- **Face shield** — *over* the goggles for any work involving acids, hot caustic, the plating bath, or passivate. The shield protects your face; the goggles are your eye backup.
- **Chemical-resistant gloves** — elbow-length, rated for acids and alkalis (nitrile, neoprene, or PVC depending on the chemistry). Inspect for pinholes before each use.
- **Chemical-resistant apron** — full-length, covering chest to below the knee.
- **Chemical-resistant boots** — closed, non-slip, splash-proof. Floors near plating tanks get wet and slick.
- **Long sleeves / appropriate clothing** — no exposed skin where splashes are likely.

⚠ **HEADS-UP**

Respiratory protection is required when ventilation is inadequate or when working over fuming tanks (acid pickle, plating bath mist, passivate). Use the cartridge or supplied-air respirator your shop specifies for that station. If you can smell acid fumes strongly, the ventilation is not keeping up — stop and report it.

💡 **TIP**

Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse. This keeps you from touching your face with contaminated gloves. And rinse your gloves *before* you take them off.

• STATION-BY-STATION HAZARD SUMMARY

STAGE	PRIMARY HAZARD	WHY IT'S DANGEROUS
01 Soak Clean	**Hot alkaline (caustic)**, 130–160°F	Chemical burns; hot liquid splashes
02 Rinse	Residual alkaline cleaner	Mild skin/eye irritant; wet/slip hazard
03 Acid Activation	**Strong mineral acid**; HCl **fumes**; flammable **H_2 gas**	Severe burns & eye damage; corrosive fumes; fire/explosion risk
04 Rinse	Dilute acid residue	Mild irritant; slip hazard
05 Acid Zinc Plating	Acidic chloride bath; **zinc chloride mist**; **high-amperage rectifier**	Burns; respiratory irritation; shock & arc-flash risk
06 Rinse	Dilute acid chloride residue	Mild irritant; slip hazard
07 Passivate	**Acidic chromate** (Cr(III) pH 1.5–3.5; or Cr(VI) where still used)	Burns; Cr(VI) is a **carcinogen**
08 Final Rinse/Dry	Hot water/air; residual chemistry	Burns from hot dryer; slip hazard

⚠ **HEADS-UP**

Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting plating chemicals — especially heavy metals like zinc, and chromium. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

CHAPTER 6

EMERGENCY & FIRST-RESPONSE

WHEN SECONDS COUNT — LEARN THIS COLD, BEFORE YOU EVER NEED IT

Know these *before* you need them. In an emergency, seconds matter and you won't have time to read. **Locate, before your first shift on the line:** the nearest **emergency eyewash** (within ~10 seconds' reach of every chemical station), the nearest **safety shower**, the **fire extinguisher** and alarm, the **rectifier emergency stop / disconnect** and main power shutoff, and the **spill kit** and **SDS binder** (Safety Data Sheets — the official hazard documents for every chemical on site).

> ⚠ **HEADS-UP**
> **Eyewash and shower stations must always be unblocked.** Never stack parts, totes, or drums in front of them. The day you need one is the day you can't move boxes out of the way.

• FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Chemical on skin	Get to the safety shower; flush **at least 15 minutes**. Remove contaminated clothing while flushing — don't stop early because "it feels better." Get medical attention; bring the SDS.
Chemical in eyes	Go to the eyewash; hold eyelids open and flush **at least 15 minutes**, rolling your eyes. Do not rub. Do not stop early. Get medical attention immediately — eye exposures are always urgent. Bring the SDS.
Inhaled fumes/mist, feel ill	Move to fresh air immediately. Report it — do not "tough it out." Seek medical attention for anything beyond brief, mild irritation. Bring the SDS.
Chemical swallowed	**Do not** induce vomiting unless the SDS directs it (vomiting acid or caustic burns the throat again). Follow SDS first aid; call medical help / poison control immediately.
Electrical shock / arc-flash	**Do not touch the person** if they may still be in contact with current. Cut power at the emergency stop / disconnect first, then call for help and render aid.
Chemical spill	Alert others and keep people away. Only clean it if trained and within your shop's "small spill" limits — otherwise evacuate and call trained spill response. Never let spills reach floor drains.

> ⚠ **HEADS-UP — LOCKOUT/TAGOUT (LOTO)**
> LOTO is mandatory for *all* rectifier and electrical maintenance. LOTO means physically locking the power off and tagging it so no one can switch it back on while someone is working on it. Never work on a rectifier that hasn't been locked out — "I'll just be a second" is how people get electrocuted.

> ⚠ **HEADS-UP — A CRITICAL ACID-MIXING RULE, FOR LIFE**
> When diluting acid, **always add ACID to WATER — never water to acid.** Adding water to concentrated acid releases heat so violently it can boil and erupt the acid out of the container into your face. The memory aid: "**Do as you oughta — add acid to water.**" Pour slowly, down the side, with the acid going *into* a larger volume of water.

> ☀ **FUN FACT**
> The reason acid-to-water matters is heat capacity: water absorbs the large heat of dilution far better when the acid is the small addition to a big pool of water. Get it backwards and the tiny bit of water flash-boils and spits acid. This one rule has saved more faces than any other in the chemical trades.

CHAPTER 7

THE SAFETY PHILOSOPHY OF AN ACID ZINC LINE

THE MINDSET THAT KEEPS YOU SAFE ACROSS A WHOLE CAREER

Rules and PPE are the *tools*. The philosophy is the *mindset* that makes them work. Five principles guide everything on this line.

1. ASSUME EVERY TANK CAN HURT YOU.

Even the rinse tanks carry residual chemistry. Even “just water” can be slippery enough to put you on the floor. Respect comes first; familiarity is where complacency — and accidents — begin. The operator who has run the line for ten years without an incident got there by *staying* careful, not by relaxing.

2. CONTAINMENT BEATS CLEANUP.

Move parts deliberately. Let them drain over the tank before transferring. Don’t fling racks, splash, or rush handoffs. Most chemical exposures aren’t dramatic spills — they’re splashes and drips from sloppy handling. Slow, smooth movement is safer *and* makes better parts (less drag-out means cleaner rinses).

3. RESPECT THE TWO FACES OF EVERY HAZARD.

On this line, the things that make good parts are the same things that can hurt you. The acid that activates the steel burns your skin. The current that deposits zinc can arc-flash. The chromate that protects the part is toxic. You can’t separate the work from the hazard — so you manage both at once, every time.

4. THE CHEMISTRY IS UNFORGIVING, SO BE DISCIPLINED.

Acid zinc punishes contamination harder than alkaline zinc. The same discipline that keeps the bath healthy — careful rinsing, no shortcuts, controlling drag-in — is the discipline that keeps *you* healthy. Good housekeeping is a safety practice, not just a quality practice.

5. WHEN IN DOUBT, STOP AND ASK.

No one expects you to have every answer, especially while learning. The unsafe operator is not the one who asks questions — it’s the one who guesses. If a fume seems strong, a glove looks worn, a reading looks wrong, or a procedure feels off, *stop and check*. That instinct is the single most valuable safety habit you can build.

• THE THREE HAZARD FAMILIES TO INTERNALIZE

1. **Acids & Alkali (corrosives).** The hot caustic cleaner (Stage 1) and the mineral acids (Stage 3) cause burns on contact and damage eyes instantly. The plating bath and passivate are acidic too. *Defense:* goggles, face shield, gloves, apron — and the 15-minute flush rule if exposed. And never forget acid-to-water.
2. **Electrical & physical.** The rectifier runs high amperage — shock and arc-flash risk. Wet floors mean slips. *Defense:* LOTO for all electrical work, never touch a rectifier or bus bar with wet hands, non-slip boots, and keep walkways clear and dry.
3. **Chromate / passivation chemistry.** Trivalent Cr(III) is an acidic irritant; hexavalent Cr(VI), if present anywhere, is a carcinogen with strict regulatory controls. *Defense:* know which your shop uses, wear the specified PPE, never let chromate mist go uncontrolled, and follow OSHA requirements to the letter if any Cr(VI) is in play.

★ **REMEMBER**

Acids and alkali, electrical and physical, chromate chemistry. Three families. If you can name the hazard family at the tank in front of you, you already know most of what you need to protect yourself.

☀ **FUN FACT**

Many veteran platers can identify a bath problem by smell or the look of a deposit before any instrument confirms it — a genuinely useful skill built over years. But notice the lesson hidden in that: *they got to be veterans by never getting hurt.* Skill and safety grow together. Build both, every shift, and you’ll have a long, healthy career in this trade.

★ **REMEMBER**

You’ve now got the whole foundation: what plating is, what makes acid zinc special, how the eight-stage line works as one connected system, why the order is locked by chemistry, what to wear, what to do in an emergency, and the safety mindset that ties it all together. Everything in the stage chapters builds on these fundamentals. Turn the page knowing the *why* — and the *how* will make a lot more sense.

PART TWO — THE PREP LINE (STATIONS 01–04)

STATION 01 OF 08

ALKALINE SOAK CLEAN

"PLATE OVER DIRT AND YOU'VE JUST PLATED DIRT."

★ **REMEMBER — THE PREP-LINE PROMISE**

By the time a part reaches the zinc tank, 80% of whether it'll turn out beautiful or scrap has already been decided. The four prep stations — Clean, Rinse, Activate, Rinse — are not the "boring stuff before the real plating." As the old-timers say: *"The plating bath gets the credit, but the pretreatment does the work."*

Welcome to the very first tank. Everything good that happens downstream starts here. Your job at this station is simple to say and important to nail: **get the part perfectly clean.** Not "looks clean." Not "pretty clean." *Chemically* clean — no oil, no grease, no fingerprints, no shop dust. If you do this one thing right, you've made every job after you easier. If you cut a corner here, that corner comes back as peeling, hazing, or spotty plating that nobody can fix later.

• WHAT YOU'RE DOING & WHY IT MATTERS

When steel parts arrive at your station, they are *not* clean — even if they look shiny. They almost always carry three layers of stuff:

1. **Particulate** — loose solid bits: shop dust, metal fines from machining, grit.
2. **Oil and grease** — the slippery film from stamping oils, cutting fluids, rust-preventives, and hand contact. You usually can't see it, but it's there.
3. **Oxide scale** — a thin layer of rust/iron oxide bonded to the steel itself. (We don't remove this one here — that's station 3's job. But it's good to know it's there.)

Soak Clean removes the first two. The cleaner is a hot, strongly alkaline (the opposite of acidic) water-based solution. It works several ways at once: **heat** softens and thins the oils so they let go of the metal; **surfactants** (the soap-like ingredients) surround tiny droplets of oil and lift them off — exactly like dish soap cutting grease off a frying pan; and **builders / alkalinity** chemically break down (saponify) certain oils, turning them into something water can carry away.

💡 **TIP — THE WATER BREAK TEST (FREE, FAST, AND NEVER LIES)**

A truly clean steel surface loves water. When you pull a clean part out, the water clings and **sheets off in one continuous, unbroken film.** A dirty surface repels water, so the water **beads up** into little droplets instead. **Continuous sheet = PASS** (send it on). **Beading or broken film = FAIL** (clean it again). The test costs nothing and takes two seconds. Use it on every rack, every time.

```
FAIL - water beads up         PASS - water sheets evenly
+--------------+              +--------------+
|  o   o   o   |  X           |##############|  OK
|    o    o    |              |##############|
+--------------+              +--------------+
Oil remains -> RE-CLEAN       Surface clean -> proceed
```

⚙ **TECHNICAL STUFF**

"Water break" refers to the *surface energy* of the metal. Clean steel has high surface energy, so water spreads to maximize contact (low contact angle). Even a one-molecule-thick film of oil drops the surface energy, so water pulls itself into beads to minimize contact (high contact angle). That's why the test can catch contamination far too thin for your eye to see.

• OPERATOR ESSENTIALS

MEMORIZE THE RANGES

PARAMETER	RANGE	WHAT TO WATCH
Temperature	130–160°F (54–71°C)	**Do not exceed 180°F**
Time	3–10 min	Heavy oil: 8–10 min · Light stamping oil: 3–5 min
Concentration	4–8 oz/gal (30–60 g/L)	Titrate (test) weekly at minimum
pH	11.0–13.5	Depends on the cleaner formulation
Agitation	Air or mechanical, continuous	Still soak for tubular parts
Filtration	Surface skimmer recommended	Removes floating oil
Water Break Test	Pass = continuous sheet	The only reliable field test

A few plain-language definitions for that table: **Concentration** = how much cleaner chemical is dissolved in the water (oz/gal = ounces per gallon); too weak and it won't cut the oil, too strong wastes money. **Titrate** = a simple lab test (your QC or lead can show you) that measures the actual concentration so you know when to add more. **Agitation** = movement in the tank (air bubbles or moving the parts) that keeps fresh cleaner contacting the surface. **Surface skimmer** = a device that pulls floating oil off the top so it doesn't redeposit on your clean parts.

⚠ **HEADS-UP — SAFETY: HOT ALKALINE SOLUTION**

This tank is **hot AND caustic.** Splashes cause **chemical burns on contact**, and the steam/mist irritates your airways. **PPE every time:** chemical splash goggles, face shield (when charging or dumping the tank), elbow-length chemical gloves, apron, chemical-resistant boots.

First aid: **skin contact → flush 15 minutes; eye contact → eyewash 15 minutes** and get help. Never lean over the tank to look in without eye protection — a stray bubble burst can reach your face.

• STEP-BY-STEP PROCEDURE

1. **Check the tank before you start.** Confirm temperature reads **130–160°F** (never above 180°F). Confirm concentration is in range (check the last titration log). Make sure the skimmer is running and the surface is reasonably free of floating oil.
2. **Inspect the load.** Heavy rust-preventive oil → plan for the long end of the time range (8–10 min). Light stamping oil → 3–5 min is plenty.
3. **Rack or load the parts** so solution reaches every surface. Avoid nesting parts where air can be trapped (trapped air = a dry spot = a dirty spot). For tubular/blind-hole parts, position them so the bore floods and drains.
4. **Immerse fully** and start your timer. Confirm agitation is working. For tubular parts, your shop may specify a still soak instead.
5. **Soak for the full time.** Don't rush it. The clock is doing chemistry you can't see.
6. **Withdraw slowly,** letting solution drain back into the tank (saves chemical, reduces drag-out).
7. **Run the water break test.** Look for a continuous sheet. **Pass → station 2. Fail → back in the cleaner.**
8. **Log it** if your shop logs cleaning, and move the rack promptly so the clean surface doesn't sit and flash-rust.

• TEACH-BACK CHECKLIST

A trainee is signed off on Station 01 only when they can, unprompted:

- [ ] Name the three kinds of contamination the soak clean removes (particulate and oil/grease — it does NOT remove oxide scale; that's station 3).
- [ ] State the temperature range and the hard ceiling (**130–160°F; never over 180°F**).
- [ ] State the time for heavy oil vs. light oil (8–10 min vs. 3–5 min).
- [ ] Correctly perform a water-break test and read the result honestly (continuous sheet = pass).
- [ ] Verbally walk through what they'd do if the water-break test fails (re-clean it — never send it forward).
- [ ] Name three pieces of required PPE.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Skipping the water break test because the part "looks fine."** Oil that ruins plating is invisible. Test every time.
2. **Running the tank too cold.** Cold cleaner can't cut oil no matter how long you soak. If the water break keeps failing, check temperature first.
3. **Letting the concentration drift low.** Cleaner gets used up. If nobody's titrating, you're slowly plating dirt without knowing it.
4. **Pulling parts through the floating oil layer.** If the skimmer is off or overwhelmed, the oil you just removed redeposits on the way out.
5. **Nesting parts so tight that air pockets form.** Wherever solution can't reach, the part stays dirty. Rack with space.
6. **Going way too hot to "clean faster."** Above 180°F you can flash off solution, etch or discolor parts, and create more mist hazard. Hotter is not better.

• QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Water break fails	Low concentration or temp	Titrate cleaner; bring temp to range
Oily film after cleaning	Floating oil redepositing	Clean/run skimmer; dump tank if heavily contaminated
White residue on parts	Hard water salts; cleaner residue	Check rinse water quality
Etching/discoloration	Cleaner too aggressive or too hot	Reduce temp; consider milder formulation
Spotty adhesion downstream	Uneven cleaning; trapped air	Reposition on rack; add agitation

SIGN-OFF — STATION 01

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I confirm the parts in this load passed the water break test (continuous sheet, no beading), the soak clean was at 130–160°F and in concentration, and I wore required PPE."

STATION 02 OF 08

RINSE (PRE-ACTIVATION)

“A RINSE ISN’T A REST STOP. IT’S A CHECKPOINT.”

You just got the part beautifully clean — but it’s now coated in a thin film of the cleaner itself. This station’s whole job is to **wash that cleaner film off** before the part goes into the acid tank. It sounds like a “nothing” step. It isn’t: a weak rinse here quietly sabotages the acid tank next door, and you won’t see the damage until parts come out wrong.

• WHAT YOU’RE DOING & WHY IT MATTERS

The soak cleaner is **alkaline** (high pH). The very next tank — acid activation — is, well, **acid** (low pH). Acid and alkaline are chemical opposites: when they meet, they neutralize each other and cancel out. So if you carry a film of alkaline cleaner straight into the acid tank, three bad things happen:

1. **You waste acid** — the acid gets used up neutralizing the cleaner instead of doing its real job on the part.
2. **You raise the acid tank’s pH** — making it weaker and less effective at activating the steel.
3. **You reduce activation quality** — which shows up later as poor adhesion (plating that peels).

This invisible carry-over of liquid from one tank to the next is called **drag-out** (what leaves the tank clinging to the part) or **drag-in** (what arrives in the next tank). The rinse station’s job is to knock that drag-out down to almost nothing — by dilution. Clean water surrounds the part and the cleaner film dissolves off into the rinse water, where it’s diluted to harmlessness and flushed away by the constant flow of fresh water.

💡 TIP — HOW WE MEASURE A CLEAN RINSE: CONDUCTIVITY

You can’t see dissolved cleaner, so we measure it with **conductivity** — how well the water conducts electricity. Pure water barely conducts at all. Dissolved chemicals make water conduct *more*. So **higher conductivity = more cleaner left in the rinse = dirtier rinse.** We measure it in **microsiemens per centimeter (µS/cm)** — just think of it as “the contamination number.” Lower is cleaner.

★ REMEMBER — THE RULE CARD NUMBER: < 500 MS/CM

Keep this rinse **below 500 µS/cm** (aim for **under 300**). Above 500, you’re dragging alkaline cleaner into the acid activation tank — wasting acid, raising its pH, and weakening activation. This number is your early-warning system. Check it daily.

• THE SMART WAY TO RINSE: COUNTER-FLOW

You *could* just blast fresh water through a single tank and dump it. It works, but wastes enormous water. The professional approach is a **counter-flow (counter-current) rinse**: the parts move forward; the water flows backward. Fresh water enters the *last* (cleanest) tank, then overflows backward into the dirtier tank. So your part gets progressively cleaner rinses, and the cleanest water it ever sees is the last thing to touch it before the acid tank.

```
FROM         +-------------+  PARTS ->  +-------------+        TO
CLEANER ->   |  RINSE 1    | ---------> |  RINSE 2    | ---->  ACID
             |  (dirty)    |           |  (clean)    |
             | <500 uS/cm  |           | aim <300    |
             +-------------+           +-------------+
                  ^   overflow <----------    ^
                  +------ <- FRESH WATER FLOWS BACKWARD <- -----+
                                                       FRESH IN
```

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient (60–85°F / 16–29°C)	No heating required
Time	30–60 sec with agitation	Longer for blind holes/tubing
Water Source	Municipal or DI	**DI preferred**
Conductivity	< 500 µS/cm (target < 300)	Monitor daily
Flow Rate	2–4 gal/min	Continuous overflow or counter-current
Tank Turnover	4–6 / hour	Prevents stagnation
pH	7–10	Elevated pH = cleaner drag-in

Plain-language definitions: **Ambient** = room temperature, no heater. **DI water** = deionized water with minerals removed (low conductivity, rinses better, no spots). **Continuous overflow** = fresh water in, dirty spills out the top. **Tank turnover** = how many times per hour the whole tank volume is replaced; 4–6 keeps it from going stale.

⚠ **HEADS-UP — SAFETY: RESIDUAL ALKALINE CLEANER**
The rinse water carries dragged-in cleaner and starts out mildly caustic. Wear **goggles, chemical gloves, and non-slip footwear** — the floor around rinses is always wet, and slips are the #1 injury here.

• STEP-BY-STEP PROCEDURE

1. **Check conductivity at shift start.** If it's near or above 500 µS/cm, tell your lead — the rinse needs more flow or a refresh.
2. **Confirm water is flowing** (continuous overflow, or counter-current cascade active).
3. **Transfer the rack promptly** from the cleaner — dried cleaner is harder to rinse off.
4. **Immerse fully with agitation, 30–60 seconds.** For blind holes and tubing, go longer and move the part to flush trapped solution.
5. **Lift and drain** over the rinse tank for a few seconds to carry less forward.
6. **Move promptly to the acid tank** — don't let the rinsed part sit and flash-rust.

• TEACH-BACK CHECKLIST

- [ ] Explain why carrying cleaner into the acid tank causes problems (neutralizes acid — wastes acid, raises pH, weakens activation).
- [ ] Name the term for chemistry clinging to a part from the previous tank (drag-out / drag-in).
- [ ] Explain what conductivity tells us and which direction is "clean" (lower = cleaner).
- [ ] State the conductivity limit and target (**< 500 µS/cm; target < 300**).
- [ ] State how long to rinse and why longer for tubing (30–60 sec; tubing traps solution).

• TOP MISTAKES NEW OPERATORS MAKE

1. **Treating the rinse as "quick."** A two-second dunk rinses nothing — give it the full 30–60 seconds.
2. **Ignoring the conductivity meter.** A high number means you're poisoning the acid tank right now.
3. **Letting the rack drip-dry between cleaner and rinse.** Dried-on cleaner is stubborn — move promptly.
4. **Not flushing blind holes and tubes,** so concentrated cleaner hides in cavities and rides forward.
5. **Skipping the drain step** — a few seconds of draining sharply cuts what you drag into the acid.

• QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
High conductivity	Heavy drag-out; low flow	More drain time off cleaner; more rinse flow; add counter-current stage
Soapy residue on parts	Inadequate rinse time	Extend dwell; add agitation
Acid tank going alkaline	Cleaner carry-over	Improve this rinse; add a second stage

SIGN-OFF — STATION 02

Trainee: ____________________ Trainer: ____________________ Conductivity: __________ µS/cm

"I confirm the pre-activation rinse conductivity was within limit (< 500 µS/cm), water flow was running, and parts were rinsed the full time with PPE worn."

STATION 03 OF 08

ACID ACTIVATION

"NO ACTIVATION, NO ADHESION. PERIOD."

This is the station that makes plating *stick*. Your clean part still has an invisible, microscopic layer of oxide (think of it as super-thin rust) bonded to the steel. Zinc can't grab onto oxide — it can only grab onto bare, chemically "awake" steel. The acid dip dissolves that oxide and leaves a fresh, reactive surface ready to bond with zinc. This is also the station with the **most safety weight** in your section — strong acid, fumes, and a hidden risk called hydrogen embrittlement.

• WHAT YOU'RE DOING & WHY IT MATTERS

Even a spotlessly cleaned steel part has a thin film of **oxide** on its surface. Oxide is what forms when iron meets oxygen — at its worst it's visible rust and scale; at its mildest it's an invisible film that forms in *minutes* on bare steel exposed to air. Either way, it's a barrier. Zinc plated on top of oxide doesn't truly bond — it sits on the oxide like paint on a greasy wall, and it will eventually flake or peel.

Acid activation (also called the **acid dip** or **pickle**) fixes this. The acid chemically dissolves the oxide layer right off the surface, leaving **clean, chemically active steel** — the bare iron atoms exposed and ready to form a strong metallurgical bond with the zinc. You'll often see tiny bubbles during the dip — that's **hydrogen gas (H_2)**, a normal sign the acid is reacting with the metal (and a clue to a hidden hazard below).

⚙ TECHNICAL STUFF

The reaction is straightforward: the acid (say HCl) reacts with iron oxide to dissolve it (iron oxide + acid → soluble iron salt + water), and once the oxide is gone the acid begins attacking the iron itself, producing hydrogen gas ($Fe + 2HCl \rightarrow FeCl_2 + H_2\uparrow$). That last reaction is why you must *time the dip* — a little attack cleans and activates; too much etches the part and floods it with hydrogen.

★ REMEMBER — THE RULE CARD: 15–60 SECONDS

The dip time is everything. **Over-dip** = you over-etch the surface (roughness, pitting) AND drive excess hydrogen into the steel. **Under-dip** = oxide remains, and your plating won't stick. Find the right time and *time it.*

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	`Ambient–110°F (Amb–43°C)`	Ambient typical; warm only for heavy scale
Time	`15–60 sec`	15–30 for clean steel · 30–60 for cast iron/heavy oxide
Acid — Option A	HCl 10–50% v/v	Most common; produces fumes — **ventilation required**
Acid — Option B	H_2SO_4 5–25% v/v	Less fuming; slightly slower; avoids adding chloride to the rinse chain
Agitation	`Manual rack movement only`	**No air agitation** — it creates acid mist
Iron Content	`Dump when > 50 g/L Fe`	Spent acid loses bite; iron can drag into the plating bath
Inhibitor	`Per vendor TDS (optional)`	Reduces base-metal attack and hydrogen absorption

Plain-language definitions: **HCl** = hydrochloric acid (also called muriatic acid). **H_2SO_4** = sulfuric acid. Both are strong mineral acids. **% v/v** = “percent by volume” — how much concentrated acid is mixed into the water (50% v/v HCl means half acid, half water). **Inhibitor** = an additive that lets the acid keep dissolving *oxide* while slowing its attack on the *base steel itself* — protecting the part and cutting hydrogen absorption. **Iron content (Fe)** = dissolved iron that builds up as the acid works; over 50 g/L the acid is “spent” — sluggish, and a contamination risk to the plating bath downstream.

⚠ **HEADS-UP — SAFETY: STRONG MINERAL ACID (READ THIS TWICE)**

HCl fumes at room temperature — you don’t have to heat it to release irritating, corrosive vapor; ventilation is required. **H_2SO_4 causes severe dehydration burns** and reacts violently with water — mixing the two releases a burst of heat. (That’s exactly why you dilute by trickling a *small* amount of acid into a *large* volume of water: the big pool of water soaks up the heat so it can’t flash to steam and spatter. Pouring water onto acid does the reverse and is dangerous.) **Hydrogen gas (H_2) released during pickling is flammable** — no open flames or sparks near the tank.

PPE: goggles + face shield, acid-resistant gloves (butyl or neoprene), apron, boots, **respirator if ventilation is inadequate.** Skin contact: flush 15 min. Eye contact: eyewash 15 min. HCl inhalation: fresh air immediately.

Spills: neutralize with soda ash. **NEVER pour water alone onto concentrated H_2SO_4** — it can boil and spatter acid. Always add acid slowly to water when diluting (“Do as you oughta — add acid to water”). **No air agitation** in this tank — bubbling air through acid throws a fine acid mist into your breathing space. Move the rack by hand instead.

THE BIG ONE: HYDROGEN EMBRITTLEMENT

⚠ **HEADS-UP — SAFETY & QUALITY: HYDROGEN EMBRITTLEMENT**

This is the most important concept in your whole section, so we'll go slow. When acid attacks steel, it releases hydrogen. Most of it bubbles away as gas — but some sneaks *into* the steel as single hydrogen atoms, wedging into the spaces between the iron atoms (the "lattice"). On ordinary mild steel this is usually harmless. But on **high-strength / hardened steel**, that trapped hydrogen makes the metal **brittle** — and a brittle part can **crack and fail suddenly under load**, sometimes hours or days later. On a critical fastener or structural part, that's a real safety failure.

The acid pickle is the single greatest source of hydrogen absorption in the entire line. Uninhibited acid on high-strength steel (**≥ 40 HRC / ≥ 180 ksi UTS**) can load the lattice with hydrogen *in seconds.*

What you do about it: Use the **minimum effective acid concentration and the minimum effective dip time** — don't over-dip "to be safe" (over-dipping is the opposite of safe here). **Always use an inhibitor on substrates over 40 HRC.** If a job is flagged as high-strength or embrittlement-sensitive, **stop and check with your lead** — alkaline zinc may be the right process instead, and hardened parts require a post-plate **bake** (handled at Station 06) to drive hydrogen back out.

Terms: **HRC** = Rockwell C hardness (higher = harder). **ksi UTS** = thousands of pounds per square inch of ultimate tensile strength. You don't have to calculate these — just recognize the words on a job traveler and treat them as a "slow down and check" flag.

STEP-BY-STEP PROCEDURE

1. **Check the job for hardness flags first.** If the traveler says high-strength, hardened, ≥ 40 HRC, or ≥ 180 ksi — pause and confirm the right process and that inhibitor is in use. Embrittlement risk comes before everything else.
2. **Confirm ventilation is running** and your PPE (including face shield) is on.
3. **Verify acid strength and the dip time** for this part type: 15–30 sec for clean mild steel; 30–60 sec for cast iron or heavy oxide/scale.
4. **Immerse the rack fully** and start your timer. **Agitate by gently moving the rack by hand** — never air.
5. **Watch the surface.** Light bubbling is normal. The dark oxide should visibly lighten/disappear. If it's not clearing, the acid may be too dilute or spent — don't just leave a hard part in longer; check the bath.
6. **Pull at the timed mark.** Don't "round up." Over-dipping etches and embrittles.
7. **Drain briefly** and move **promptly** to the pre-plate rinse (station 4) — activated steel re-oxidizes fast in air.
8. **Watch the iron level over time.** When the bath hits **> 50 g/L Fe** (or your shop's limit), flag it for dumping/recharge.

TEACH-BACK CHECKLIST

- [ ] Explain what the acid dip removes and why it matters (the oxide layer — zinc can't bond to oxide, only to bare active steel).
- [ ] Explain what happens if you over-dip vs. under-dip (over: etching + embrittlement; under: oxide remains, plating peels).
- [ ] State the dip time for clean steel vs. heavy oxide (15–30 sec vs. 30–60 sec).
- [ ] Explain why there is NO air agitation in this tank (air creates a hazardous acid mist).
- [ ] Explain hydrogen embrittlement and which parts are at risk (≥ 40 HRC / ≥ 180 ksi).
- [ ] Name the two things that reduce embrittlement risk here (minimum dip time/concentration, and inhibitor on hard parts).
- [ ] Recite the rule for diluting acid (always add acid to water, never water to acid).

TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Dark oxide remains	Acid too dilute; time too short	Increase concentration; extend time *(check embrittlement risk first)*
Pitting / etching	Acid too strong, too long, or too warm	Reduce concentration; shorten dip; add inhibitor
Adhesion failure downstream	Incomplete activation; residual oxide	Re-check acid strength/time; check for alkaline drag-in from station 2
Excessive HCl fuming	High concentration/temp; poor ventilation	Reduce to minimum effective; check exhaust
Dark smut on surface	High dissolved iron; copper contamination	Dump and recharge acid; investigate copper source
H-embrittlement failures	Over-long pickle; no inhibitor; hard steel	Shorten pickle; add inhibitor; review hardness; consider alkaline zinc

SIGN-OFF — STATION 03

Trainee: ____________________ Trainer: ____________________ Dip used: ________ sec · Inhibitor? Y / N

"I confirm I checked the job for hardness flags, ran the acid dip at the timed interval (15–60 sec) with manual agitation only, ventilation and PPE in place, and used inhibitor where required."

STATION 04 OF 08

RINSE (PRE-PLATE)

THE GUARD RINSE — THIS RINSE PROTECTS THE MOST PRODUCTIVE TANK ON THE LINE

You're at the last station before the zinc plating bath — the tank that actually makes the company money. Your job here is to **wash the acid (and the dissolved iron it carries) off the part** so it doesn't contaminate that expensive, sensitive plating bath. Think of this rinse as the **guard at the door** of the most valuable room in the building. Everything you've done in stations 1–3 comes down to handing the plating tank a clean, activated, acid-free part. Let's not fumble it at the goal line.

• WHAT YOU'RE DOING & WHY IT MATTERS

Coming out of the acid dip, the part is wet with acid — and that acid now contains **dissolved iron** it picked up while pickling. If you carry that into the plating bath, you cause three specific problems:

1. **You drop the plating bath's pH.** The acid zinc bath runs in a tight, slightly-acidic window (pH 4.8–5.8). Dragging in extra acid pushes the pH down, which throws off the deposit (burning at high-current areas) and forces constant chemical correction.
2. **You break down the brighteners faster.** The plating bath uses a delicate brightener system (the additives that make zinc come out shiny). Excess acid drag-in accelerates their breakdown — meaning duller parts and more money spent on additives.
3. **You contaminate the bath with iron.** This is the big one, and it's unique to acid zinc…

★

REMEMBER — IRON IS A ONE-WAY STREET IN ACID ZINC

In the older *alkaline* zinc process, any iron that sneaks in simply precipitates out as a solid (iron hydroxide) and gets caught by the filter — the bath cleans itself. **Acid zinc does NOT do this.** Iron that drags in **stays dissolved and accumulates** in the bath, building up over time and causing haze and dark deposits. There's no easy "filter it out." **This pre-plate rinse is your primary — and main — defense against iron contamination.** That's why we hold it to a tighter standard than any rinse before it.

CONTAMINANT	SOURCE	EFFECT ON THE ZINC BATH	PREVENTION
Iron (Fe)	Spent pickle	Haze, dark deposits; accumulates — can't be filtered out	Refresh pickle; double rinse; hold conductivity tight
Free acid	HCl/H_2SO_4 drag-out	Drops bath pH; forces constant correction	Drain well; tighter rinse
Acid + brightener load	Excess drag-in	Accelerates brightener breakdown	Add a second rinse stage

💡

TIP — CONDUCTIVITY, TIGHTER THIS TIME: < 200 MS/CM

Same measurement as station 2 (the "contamination number"), but here the limit is **under 200 µS/cm, with a target under 100.** Tighter, because acid and iron drag-in directly shorten the life of the plating bath. Check it daily and don't let it creep.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	`Ambient (60–85°F / 16–29°C)`	No heating required
Time	`30–60 sec with agitation`	**Double rinse recommended**
Water Source	`DI preferred`	Minimizes contamination to plating bath
Conductivity	`< 200 µS/cm (target < 100)`	Critical for acid zinc bath life
pH	`4.0–8.0`	Below 4 = excessive acid drag-in
Flow Rate	`Continuous overflow, 3–5 gal/min`	Counter-current ideal
Drain Time	`15–20 sec minimum`	Prevents diluting the plating bath

Plain-language notes: **Double rinse** = two rinse tanks in a row. The first knocks off most of the acid; the second polishes it down to the tight target. Strongly recommended here because of how sensitive the plating bath is. **Drain time** matters more here than anywhere: this is the last drain before the money tank. A solid 15–20 seconds means less water (and less leftover contamination) goes into the plating bath. **pH below 4** in the rinse is a warning you're dragging in too much acid — tighten up upstream and add rinsing.

> ⚠ **HEADS-UP — SAFETY: DILUTE ACID RESIDUE**
> The water here carries dragged-in acid. It's diluted, but still irritating. Wear **goggles, chemical gloves, and non-slip footwear.** And as always around rinses — the floor is wet; watch your footing.

THE 30-SECOND CLOCK: GET TO ZINC FAST

> ★ **REMEMBER**
> **Move into the plating bath within 30 seconds** of leaving the rinse. A freshly activated, rinsed surface starts to oxidize in air almost immediately — and an oxidized surface plates poorly, undoing station 3's work. Keep the part wet. Speed matters at this handoff.

STEP-BY-STEP PROCEDURE

1. **Check the conductivity** at shift start. If it's above 200 µS/cm, the rinse needs more flow or a refresh before you plate through it.
2. **Confirm water flow** (continuous overflow at 3–5 gal/min; counter-current cascade running if equipped).
3. **Transfer promptly from the acid tank** — don't let the activated, acid-wet part sit and re-oxidize.
4. **Immerse fully with agitation, 30–60 seconds.** If you have a double rinse, give both tanks their time.
5. **Drain a full 15–20 seconds** over the rinse before moving on. This step is non-negotiable here.
6. **Move into the plating bath within 30 seconds** of leaving the rinse.

TEACH-BACK CHECKLIST

- [ ] Name the three problems acid drag-in causes in the plating bath (drops pH; breaks down brighteners faster; adds iron).
- [ ] Explain why iron is a bigger deal in acid zinc than alkaline zinc (alkaline precipitates and filters it out; acid zinc accumulates it — this rinse is the main defense).
- [ ] State the conductivity limit here and how it compares to station 2 (< 200 µS/cm, target < 100 — tighter than station 2's < 500).
- [ ] State how long to drain before moving on, and why (15–20 sec; less drag-in dilutes/contaminates the plating bath less).
- [ ] State how fast the part must reach the plating tank after rinsing (within 30 seconds, before it re-oxidizes).

TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Conductivity high	Heavy acid drag-out; low flow	Increase drain time from pickle; increase rinse flow
Plating bath pH dropping fast	Acid drag-in from pickle	Tighten conductivity; add a double rinse
Iron climbing in plating bath	Dissolved Fe dragging in from spent pickle	Dump/refresh the pickle; improve this rinse; add a second stage
Parts oxidizing before plating	Transfer too slow; part drying out	Speed up — enter plating within 30 sec of leaving the rinse

SIGN-OFF — STATION 04

Trainee: ____________________ Trainer: ____________________ Conductivity: __________ µS/cm

"I confirm the pre-plate rinse conductivity was within limit (< 200 µS/cm), parts were rinsed and drained the full time, and moved to plating within 30 seconds with PPE worn."

> ★ **REMEMBER — THE PREP-LINE PROMISE**
> Four stations, one mission: hand the zinc bath a surface that is **clean, contaminant-free, oxide-free, and still wet.** Clean it (01), rinse the cleaner off (02), strip the oxide (03), rinse the acid and iron off (04), and run to the tank. **Clean → rinse → activate → rinse.** Every one a checkpoint, not a rest stop. Do these four right, every rack — and bright zinc will reward you. Cut corners here, and no amount of bath chemistry downstream can save the job.

PART THREE — THE MAIN LINE (STATIONS 05–07)
STATION 05 OF 08 · THE MAIN TANK

ACID ZINC PLATING

WHERE BARE STEEL BECOMES PROTECTED STEEL

Take a breath. Up to now, every station — cleaning, rinsing, acid activation, rinsing again — has had one job: get the steel ready. None of those tanks put any zinc on the part. This tank is the payoff. **Station 05 is where the zinc coating is actually created.** Everything before it was preparation; everything after it is protection and finishing. If you only deeply understand one station on the whole line, make it this one.

Here's the big idea in one sentence: **we use electricity to pull dissolved zinc out of a liquid and stick it, atom by atom, onto your steel part.** Let's meet the cast of characters:

- **The cathode** — that's *your part.* "Cathode" is the technical name for the negatively-charged side. **Cathode = the part = negative = where zinc lands.**
- **The anode** — bars or balls of nearly-pure zinc hanging in the tank, on the positive side. As we plate, they dissolve and *feed fresh zinc back into the liquid.*
- **The electrolyte** — the bath liquid itself: water loaded with dissolved zinc and salts that carry electricity. The highway the zinc travels on.

☀ **FUN FACT**

Because the anodes dissolve to replace the zinc that plates out, an acid zinc bath is almost self-feeding for metal. As long as you keep enough zinc anode in the tank, the zinc concentration stays remarkably stable. You're mostly topping off chemistry and additives, not metal.

WHY ACID CHLORIDE ZINC? (AND WHY IT'S LESS FORGIVING)

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
High cathode efficiency (typically ~95–98%, supplier/condition-dependent)	Almost every bit of electricity you pay for actually deposits zinc
Fast plating	2–3 times faster than alkaline zinc at the same current
Bright right out of the tank	With the right additives, parts come out shiny — no extra polishing
Runs cool	Often ambient to 95°F — less energy spent heating tanks
No cyanide	Safer, simpler waste treatment than the old cyanide formulas

★ **REMEMBER**

The headline number for this tank is **95–98% cathode efficiency.** Nearly every amp-hour of electricity becomes zinc on the part. As the tagline says: *"95% efficiency means 95% of your current is making money. Control the other 5%."*

That same high efficiency means **the bath has almost no tolerance for contamination.** Acid zinc *punishes* sloppiness: **poor throwing power** (it covers deep recesses noticeably worse than alkaline zinc — the surface-to-recess thickness ratio is roughly 3:1 to 5:1 vs. alkaline's 1.5:1 to 2:1, and the number closer to 1:1 is the more even one); **sensitivity to metal contamination** (iron, copper, lead ruin deposits); **complex three-part brightener chemistry**; **higher hydrogen-embrittlement risk** on hard steel; and it's **corrosive to equipment** (chloride eats mild steel, so tanks and fittings are plastic).

⚠ **HEADS-UP (SAFETY & EQUIPMENT)**

Never let bare mild steel touch this bath — not a tool, not a dropped fixture, not a tank repair. Chloride attacks it, and the dissolved iron then contaminates your bath. Use only plastic or properly coated equipment in and around this tank.

• THE BATH RECIPE: WHAT'S IN THE TANK AND WHY

The acid zinc bath is a carefully balanced recipe of four kinds of ingredients. Know what each one *does*, and you'll know what goes wrong when it drifts.

1. ZINC (THE METAL THAT DOES THE COATING)

Zinc dissolved in the bath, supplied mainly as **zinc chloride ($ZnCl_2$). Rack:** 15–45 g/L · **Barrel:** 20–45 g/L. It *is* the coating — no zinc, no plating. Too low: burning at high-current areas (edges, tips) — dark, rough, powdery. Too high: wastes chemistry and dulls the high-current brightness.

2. POTASSIUM OR AMMONIUM CHLORIDE (THE CONDUCTOR)

A salt — either **potassium chloride (KCl)** or **ammonium chloride (NH_4Cl)** — that makes the bath conduct electricity well. **Rack:** 120–250 g/L · **Barrel:** 150–250 g/L. Three jobs: carries current efficiently, helps keep zinc in solution, and keeps the anodes dissolving smoothly instead of going passive.

> ⚙ **TECHNICAL STUFF — KCL VS NH_4CL, THE FAMILY FEUD**
>
> Both work. The difference is downstream in waste treatment. Ammonium chloride baths release **ammonia** into the wastewater, which is a regulatory headache and needs an extra removal step. Potassium chloride avoids the ammonia entirely. KCl costs a bit more up front and plates slightly differently, but the industry is steadily moving toward KCl to stay out of trouble with ammonia in effluent. **Know which one your shop runs — they are NOT interchangeable mid-bath.**

3. BORIC ACID (THE PH BODYGUARD)

Boric acid (H_3BO_3) — a mild acid that acts as a *buffer* (a chemical shock-absorber for pH). **Rack:** 20–35 g/L · **Barrel:** 25–35 g/L. Right at the part's surface, the chemistry tries to swing the local pH upward as plating happens; boric acid holds the line in that thin film, keeping the deposit smooth and bright. It's the most-neglected ingredient in the tank because nothing dramatic happens the *day* it runs low — but deposits slowly get rough and hazy.

> ★ **REMEMBER**
>
> Do **not** neglect boric acid. It's quiet but essential. When deposits start going subtly rough or hazy and the brighteners check out fine, boric acid is a prime suspect. Have it analyzed on your regular schedule.

4. THE BRIGHTENER SYSTEM (THE COSMETIC CHEMISTRY)

Three separate additives working as a team to make the deposit bright, smooth, and fine-grained, dosed *per your chemistry supplier's TDS.* Full section on the next page.

AND THE NUMBER THAT GOVERNS ALL OF IT: PH

Target pH: 4.8–5.8 for both rack and barrel — mildly acidic. Add **hydrochloric acid (HCl)** to bring pH down; add **potassium hydroxide (KOH)** to bring it up.

> 💡 **TIP**
>
> A simple memory hook for the four big bath numbers: **Zinc 15–45, salt 120–250, boric 20–35, pH 4.8–5.8.** Keep those four in your head and you can sanity-check any analysis sheet at a glance.

• THE BRIGHTENER SYSTEM — THREE PLAYERS, ONE TEAM

There are **three brightener components**, and each controls a different part of the deposit's appearance. They are dosed *per TDS* and replenished based on how much you've plated (amp-hours), not on a clock.

COMPONENT	PLAIN-LANGUAGE JOB	IF LOW	IF TOO MUCH
Carrier (conditioner)	The foundation. Refines grain, prevents pits, helps the bath "wet" the part	Pitting, roughness, haze	Haze, *less* brightness, foaming
Primary brightener	The main shine, across the whole part	Dull deposit, worst in mid-current areas	Brittle, peeling, high internal stress
Secondary brightener	The shine in high-current areas (edges, tips) plus leveling	Dull edges and tips	Burning at low-current areas, brittle edges

The key insight: **brightness problems have a location.** *Where* on the part it looks bad tells you *which* brightener is off. Dull in the middle? Look at primary. Dull only on the edges and points? Look at secondary. Pitted or rough all over? Look at carrier.

★ **REMEMBER**

Carrier is the foundation; primary is the overall shine; secondary is the edge-and-tip shine. More of any one is NOT better — every component has a "too high" failure mode just as real as its "too low" one.

⚙ **TECHNICAL STUFF**

Why dose by amp-hours instead of by time? Because brighteners get *consumed* by the plating reaction. The more zinc you've plated (measured in amp-hours — amps times hours of current), the more brightener you've used up. A tank running hard all day burns brightener faster than one sitting idle. Replenishing per amp-hour keeps the chemistry steady regardless of how busy the shift was.

• YOUR BEST FRIEND: THE HULL CELL

Before you ever dump chemistry into a production tank based on a hunch, you test your theory in a tiny tank first — the **Hull Cell**, a small, wedge-shaped test tank. You plate one little panel in it under standard conditions — commonly **267 mL of bath at 2 amps for 10 minutes**, though the exact current and time are supplier/condition-dependent (follow your TDS). Because the cell is wedge-shaped, the panel sees *high current at one end and low current at the other all at once*. So a single panel shows you the whole current-density range in one shot — like an X-ray of your bath. Read it against your supplier's reference panels, since "good" appearance is product-dependent.

WHAT THE PANEL LOOKS LIKE	WHAT IT'S TELLING YOU
Bright across the whole panel	Bath is in balance — you're good
Dull in the middle, bright at high-CD end	Low **primary** brightener
Dull only at the high-CD end	Low **secondary** brightener
Hazy all over	Organic OR metallic contamination
Dark / burnt at the high-CD end	pH too low, or zinc too low
Pitting across the panel	Low carrier, or poor filtration

💡 **TIP**

Run a Hull Cell *before* and *after* any chemistry addition. Before, to diagnose. After, to confirm the fix worked — on a 267 mL panel instead of risking your whole production tank. It's the cheapest insurance on the line.

• METALLIC CONTAMINATION: THE ACHILLES' HEEL

This bath punishes contamination — specifically, **dissolved metals that don't belong.** Three are especially nasty, and the action levels are shockingly small.

CONTAMINANT	ACTION LEVEL	WHAT IT DOES TO YOUR PARTS	WHERE IT SNEAKS IN FROM
Iron (Fe)	`10 ppm`	Haze, dark deposits	Pickle drag-in; corroded steel equipment in the bath
Copper (Cu)	`0.5 ppm`	Black spots, adhesion failure	Brass fittings; corroded bus bars
Lead (Pb)	`0.5 ppm`	Dark deposits, dendrites (spiky growths)	Impure anodes (less than 99.99% pure)

⚠ **DON'T CONFUSE YOUR IRON NUMBERS**

This **10 ppm** is the iron limit for the *plating bath* — far tighter than the **50 g/L** dump limit you learned for the *pickle* at Station 03. Two different tanks, two very different numbers (10 *parts per million* here vs. 50 *grams per liter* there). The pickle can hold a lot of iron and keep working; the plating bath cannot tolerate almost any.

★ **REMEMBER**

"ppm" means **parts per million.** A copper limit of 0.5 ppm is half a gram of copper in a thousand liters of bath. That's *tiny.* This is why we obsess over rinses, why anodes must be 99.99% pure, and why no brass or bare steel goes near the tank.

⚙ **TECHNICAL STUFF — WHY IRON IS SNEAKIER IN ACID ZINC**

In an *alkaline* zinc bath, dissolved iron quietly precipitates out as a solid (iron hydroxide) and gets caught by the filter. In *our acid* bath, iron stays dissolved and just keeps accumulating — there's no self-cleaning. That's why your pre-plate rinse (Station 04) and a healthy pickle tank matter so much; they're your only real defense against iron creeping up.

How do you get rid of metallic contamination? Two common tools: **Dummying** — plating onto a scrap "dummy" cathode at low current to selectively pull contaminating metals out of solution (sacrificing a junk panel to clean the bath). And **carbon treatment** — circulating the bath through activated carbon to adsorb organic junk and some metals, often combined with thorough filtration.

One special case — chromium (Cr^{6+}) drag-back. Hexavalent chromium that drifts *backward* from the passivate into the zinc bath causes blistering and bare/skip-plating, and dummying alone won't clear it. The standard removal is a **reduce-then-precipitate** step: reduce the Cr^{6+} to Cr^{3+} with a reducing agent (per your supplier's TDS), **raise the pH to precipitate the chromium as an insoluble hydroxide,** then **filter it out** and re-adjust the bath pH back to 4.8–5.8. Do this only under your chemist's direction.

• THE SUPPORTING EQUIPMENT (AND WHY EACH PIECE EXISTS)

EQUIPMENT	SETTING / SPEC	WHY IT'S THERE
Anodes	`Zinc 99.99% SHG, dissolving type`	The self-replenishing zinc supply. Purity matters — cheap anodes carry lead
Anode bags	`Polypropylene, always`	Catch the fine sludge anodes shed, so it doesn't cause roughness
Agitation	`Air + cathode-rod (rack); barrel 6–12 RPM`	Keeps fresh chemistry at the surface; prevents gas pockets & uneven plating
Filtration	`Continuous, 2–4 turnovers/hr; carbon canister`	Constantly removes particles. "Turnover" = a full tank-volume through the filter
Temperature	`70–95°F (21–35°C), target 80–85°F`	Too cold = dull; too hot = brightener breakdown. Often needs *cooling*
Current density	`1–6 A/dm² (10–60 ASF) rack; 0.3–1.5 A/dm² barrel`	How hard you're driving. Too high in any spot = burning

⚙ **TECHNICAL STUFF**

"Current density" is the amount of current spread over the part's surface area (amps per square decimeter, A/dm², or amps per square foot, ASF). It's *not* total amps — it's amps per unit of area. Edges and points naturally see higher current density than flat faces, which is exactly why edges burn first and why secondary brightener exists to protect them.

ACID ZINC TANK CROSS-SECTION — TANK ANATOMY

• STANDARD OPERATING PROCEDURE (STATION 05)

1. **Confirm the part is clean and active.** It should have come straight from the pre-plate rinse (Station 04) and entered plating **within ~30 seconds** of leaving that rinse. A part that sat and oxidized will plate poorly. If in doubt, send it back upstream.
2. **Check bath readiness before the load goes in:** temperature 70–95°F (target 80–85°F); pH 4.8–5.8 (HCl down / KOH up); agitation running (air + cathode rod, or barrel turning 6–12 RPM); filtration running; anodes present, bagged, and not depleted below the solution line.
3. **Verify your chemistry is current** — zinc, chloride salt, boric acid, and brighteners all analyzed and replenished on schedule (per TDS / amp-hours).
4. **Set the rectifier** to the correct current for the part's surface area and target current density.
5. **Load the part / start the cycle.** Typical plating time runs from a few minutes to ~40 minutes depending on required thickness.
6. **Monitor while running** — watch for unusual color, gassing, or noise; confirm agitation and filtration stay on.
7. **At cycle end, remove and drain over the tank,** then transfer promptly to the post-plate rinse (Station 06).
8. **Log it** — amp-hours run, additions made, temperature, pH. This log is how you dose brightener correctly tomorrow.

⚠ **HEADS-UP (HYDROGEN EMBRITTLEMENT — READ THIS TWICE)**

Because acid zinc runs at high efficiency, more atomic hydrogen gets absorbed into the steel per amp-hour than in lower-efficiency baths. On **hardened steel (above 40 HRC / ~180 ksi)**, that hydrogen can make parts crack later without warning. Such parts must be **baked** after plating (covered at Station 06). For embrittlement-sensitive, high-strength parts, the shop should consider whether alkaline zinc is the better choice — flag it, don't assume.

COMMON DEFECTS — WHAT YOU'LL SEE AND WHAT IT MEANS

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Burning at high CD (dark, rough edges/tips)	pH too low; zinc too low; excess secondary brightener	Raise pH; add zinc; run a Hull Cell
Dull deposit overall	Primary brightener depleted; temp too low; organics	Add primary per amp-hour; check temp; carbon treat
Pitting across the part	Low carrier; poor agitation; dirty parts	Add carrier; increase air agitation; check cleaning
Hazy deposit	Iron contamination; organics; excess carrier	Analyze Fe; carbon treat; check carrier level; dummy
Dark streaks or spots	Copper or lead contamination; anode sludge	Analyze metals; dummy plate; check anode bags
Roughness	Particulate; lead; poor filtration	Check/replace filter; analyze Pb; bag anodes
Peeling / poor adhesion	Excessive internal stress; brightener imbalance	Hull Cell bend test; reduce primary; analyze Cu
Poor coverage in recesses	Normal for acid zinc (throwing power 3:1–5:1); pH too low	Optimize pH; accept some variation vs alkaline

SAFETY — THIS STATION DEMANDS YOUR RESPECT

⚠ HEADS-UP — ACIDIC CHLORIDE BATH & MIST

The bath runs acidic (pH 4.8–5.8) and full of chloride. Air agitation throws up a fine **zinc chloride mist** you must not breathe — local exhaust ventilation is mandatory. Wear full PPE: splash goggles *and* face shield, elbow-length chemical-resistant gloves, apron, boots. If you get bath on your skin, flush 15 minutes and report it.

⚠ HEADS-UP — THE RECTIFIER IS HIGH AMPERAGE

That power supply pushes a *lot* of current — shock and arc-flash risk. Before any maintenance, reaching into the tank, or work on bus bars — **Lock Out / Tag Out (LOTO)** the rectifier. Physically lock the power off and tag it. No exceptions, ever, no matter how quick the job seems.

⚠ HEADS-UP — RIGHT TANK, RIGHT METAL

Chloride solution corrodes mild steel. These tanks, heaters, and fittings are **plastic (PP/PVC)** for a reason. Don't drop steel tools or hardware into the bath — they corrode, dump iron contamination (which can't be filtered out), and can short across electrodes.

TEACH-BACK & TOP MISTAKES — STATION 05

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part — and which charge?
- The four target numbers: zinc, salt, boric, pH?
- Boric acid's job, and why it's the "quiet" ingredient?
- The three brighteners and what each controls?
- Dull only on edges/tips — which brightener?
- The three standard Hull Cell conditions?
- Why does iron accumulate here but not in alkaline?
- Why must anodes be 99.99% pure and bagged?
- Why is HE a bigger concern in high-efficiency acid zinc?

TOP MISTAKES TO AVOID

- Letting boric acid drift low (invisible until rough).
- Chasing brightness by dumping in more brightener.
- Ignoring rising iron — it only goes up here.
- Using non-99.99% anodes (introduces lead).
- Running too hot (breaks down brighteners).
- Skipping anode bags (sludge → roughness).
- Plating a part that sat too long and oxidized.

SIGN-OFF — STATION 05

I confirm that I can independently: verify bath readiness (temp, pH, agitation, filtration, anodes); read an analysis sheet against target ranges; run and interpret a Hull Cell panel; identify which brightener is responsible for a given appearance defect; recognize the three metallic contaminants and their sources; and state the hydrogen-embrittlement precaution for hardened steel. I understand that brightener additions and chemistry adjustments are made only by authorized personnel.

Operator: ____________________ Date: __________ Trainer: ____________________ Date: __________

STATION 06 OF 08

POST-PLATE RINSE & THE BAKE DECISION

WASH OFF THE CHLORIDE BEFORE IT POISONS THE FINISH — AND DECIDE WHETHER A PART GOES TO THE OVEN

Your part just came out of a chloride-loaded plating bath, and it's dripping with that chloride. The very next tank is the passivate (Station 07). Chloride is *poison* to that passivate. As the saying goes: *"Chloride into chromate is a slow death for the passivate bath."* This rinse is the wall between the two. Skip it or run it sloppy, and you slowly kill your most finicky chemistry tank downstream. It's also the decision point for whether a hardened-steel part goes to an *oven* before anything else.

> ★ **REMEMBER**
> A rinse is never "just a rinse." Each one is a *checkpoint* that protects the next tank. This one protects the passivate from chloride.

• WHAT AND WHY

When your freshly plated part is pulled from Station 05, a thin film of acid zinc bath clings to it — full of **chloride** (the "Cl" in zinc chloride and potassium chloride). If that chloride rides into the passivate tank — **drag-out** becoming **drag-in** — three bad things happen: (1) the passivate bath **degrades faster**; (2) conversion coatings come out **thin or blotchy**; (3) passivate **bath life drops** — your single most expensive recurring problem. The post-plate rinse's whole job is to wash that chloride film off *before* it reaches the passivate.

• OPERATOR ESSENTIALS

PARAMETER	TARGET	WHY IT'S SET THERE
Conductivity	`< 200 µS/cm (aim < 100)`	The headline number. Chloride drag-in is the whole concern
Temperature	`Ambient`	No heating needed
Time	`30–60 sec with agitation`	Long enough to flush the film; **double rinse strongly recommended**
Drain time	`15–20 sec over the tank`	Excess water carried forward dilutes the passivate
Water	`DI preferred`	Avoids hard-water haze on fresh zinc

> 💡 **TIP — THE MAGIC OF THE DOUBLE COUNTER-CURRENT RINSE**
> Instead of one rinse tank, use two in series. Parts move *forward* (dirty → clean), while fresh water flows *backward* (clean → dirty) over a weir. Parts get their final rinse in the cleanest water, while you use far less water overall.

```
FROM         +-------------+  PARTS ->  +-------------+        TO
PLATING ->   |  RINSE 1    | ---------> |  RINSE 2    | ---->  PASSIVATE
             |  (dirty)    |           |  (clean)    |
             | 800+ uS/cm  |           | <200 uS/cm  |
             +-------------+           +-------------+
                  ^   overflow <----------    ^
                  +------ <- FRESH WATER FLOWS BACKWARD <- -----+
                                                      FRESH IN
```

• STANDARD OPERATING PROCEDURE

1. **Transfer promptly from plating.** Don't let the plated part sit and oxidize in air.
2. **Immerse with agitation, 30–60 seconds.** Make sure blind holes and recesses actually flush — trapped pockets hold chloride.
3. **If double-rinsing,** move dirty → clean so the final contact is the cleanest water.
4. **Drain 15–20 seconds** over the tank before moving on, so you don't carry a flood of water into the passivate.
5. **Handle with clean gloves only** — bare fingers leave oils and salts that show up as fingerprints and contamination on the bright, fresh zinc.
6. **Move promptly to passivate** (Station 07) before the zinc starts to oxidize in air.

RINSE QUALITY MONITORING (DAILY CHECKS)

CHECK	METHOD	HOW OFTEN	TAKE ACTION WHEN…
Conductivity	Meter	Every load	**> 200 μS/cm** → increase fresh-water flow
pH	Strip / meter	2× daily	**< 5.0** → check for acid/chloride drag-in
Temperature	Thermometer	Daily	**> 90°F** → check cooling
Visual clarity	Observe	Every load	Cloudy → dump and refill
Zinc content	Titration	Weekly	**> 50 ppm** → increase overflow

THE HYDROGEN-EMBRITTLEMENT BAKE (CRITICAL FOR HARD STEEL)

This is the natural point in the line to talk about **baking**, because some specs require it *before* passivation. High-strength steel absorbs atomic hydrogen during pickling and high-efficiency plating, and that hydrogen can cause delayed cracking. **Baking** drives it back out of the steel before it can do damage.

> ⚠ **HEADS-UP (SAFETY-CRITICAL — THIS PREVENTS CATASTROPHIC PART FAILURE)**
> For hardened steel **above 40 HRC**: bake **within 4 hours** of the last electroplating step, at **375 ± 25°F**, for a **minimum of 8 hours**, per **ASTM B850.** Some customer specs require the bake *before* passivation — check the spec. A missed bake on a structural fastener isn't a cosmetic defect; it's a part that can snap in service. Never skip it on embrittlement-sensitive work.

> ⚙ **TECHNICAL STUFF**
> "HRC" is the Rockwell C hardness scale. The higher the HRC, the harder — and the more brittle and embrittlement-prone — the steel. The 4-hour window matters because the longer hydrogen sits trapped in stressed steel, the more damage it can start before you bake it out.

TROUBLESHOOTING (STATION 06)

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Passivate bath life is short	Chloride drag-in from this rinse	Tighten conductivity; add a second rinse stage
Blotchy passivate	Residual chloride still on the zinc	Improve rinse agitation and increase immersion time
White haze on the zinc	Zinc oxidizing, or hard-water deposits	Use DI water; reduce transfer time between tanks

TOP MISTAKES AT STATION 06

1. **Treating it as optional.** This rinse directly determines passivate bath life and finish quality.
2. **Letting conductivity climb above 200 μS/cm** without bumping fresh-water flow.
3. **No drain time** — flooding the passivate with carried-over water and diluting it.
4. **Bare-handed handling** of fresh zinc, leaving fingerprints and contamination.
5. **Skipping or delaying the embrittlement bake** on hardened steel.

TEACH-BACK CHECKLIST

- [ ] Explain why chloride drag-in matters so much at *this* particular rinse.
- [ ] State the conductivity target, and what μS/cm actually measures.
- [ ] Explain a double counter-current rinse: which way parts move, which way water flows, and why it saves water.
- [ ] State the embrittlement bake parameters exactly (temperature, time, time-window, HRC threshold, standard).
- [ ] Explain why fresh zinc must be handled with clean gloves.

SIGN-OFF — STATION 06

I confirm that I can independently: measure and interpret rinse conductivity; maintain the < 200 μS/cm target and respond when it's exceeded; run a double counter-current rinse correctly; state and apply the ASTM B850 bake requirement for hardened steel (375 ± 25°F, minimum 8 hours, within 4 hours of plating); and handle fresh zinc without contaminating it.

Operator: ______________________ Date: ____________ Trainer: ______________________ Date: ____________

STATION 07 OF 08 (+ STAGE 8 FINAL RINSE/DRY)

PASSIVATE — TRIVALENT CHROMATE

THE CHROMATE CONVERSION COATING THAT TURNS BARE ZINC INTO REAL PROTECTION

You've made bright zinc — surely that's the protection? The zinc *is* protective, but bare zinc is **reactive.** Left alone, it quickly grows a powdery white corrosion product everyone calls **white rust** (zinc oxide/hydroxide). It looks terrible and shortens the coating's protective life. So we add one more coating *on top of* the zinc: a **chromate conversion coating**. As the tagline says: *"Zinc without passivate is a coating waiting to white rust."*

★ **REMEMBER**

The zinc protects the steel. The passivate protects the zinc. It's protection on protection — and the last station is where that final shield goes on.

WHAT "CONVERSION COATING" MEANS (AND WHY IT'S DIFFERENT FROM PLATING)

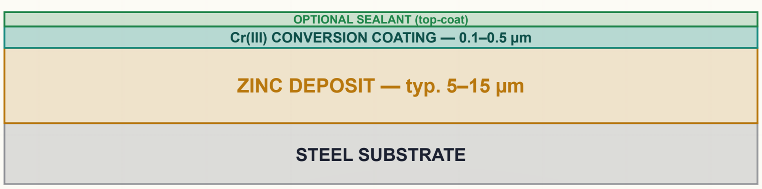
Station 05 was **electroplating** — electricity *added* a separate layer of metal. Station 07 is a **conversion coating** — *no electricity.* You simply dip the zinc-plated part into the passivate solution, and the solution chemically *reacts with the surface of the zinc itself,* converting the very top layer of zinc into a thin, tough, protective film. The film isn't a separate thing laid on top; it's the zinc surface *transformed.* That's why it's called "conversion."

⚙ **TECHNICAL STUFF**

In the dip, the mild acid in the passivate lightly dissolves the outermost zinc, and chromium compounds in the bath deposit into that reacting surface, building a gel-like film rich in chromium that then dries hard. The result is a stable barrier layer that slows corrosion. (Older *hexavalent* Cr(VI) films could "self-heal" minor scratches because they released soluble chromium that re-passivated bare zinc; modern *trivalent* Cr(III) films generally cannot self-heal — they protect purely as a stable barrier.) The cross-section below shows the stack: steel substrate, zinc deposit (typically 5–15 µm), then the thin Cr(III) conversion coating (0.1–0.5 µm), with an optional sealant on top.

PASSIVATE CROSS-SECTION (NOT TO SCALE)

CONVERSION = THE ZINC SURFACE TRANSFORMED



THE COLORS: CLEAR/BLUE VS YELLOW VS BLACK

Passivates come in colors. The color generally signals **how thick the film is** and **how much corrosion protection** you get — thicker chromate films are both more protective and more colored.

TYPE	LOOK	RELATIVE PROTECTION
Clear / Blue	Nearly clear, faint bluish	Good — entry-level protection
Yellow / Iridescent	Yellow with a rainbow shimmer	Highest of the common trivalent types
Black	Black	Decorative + protective

💡 **TIP**

The customer's spec decides the color, not operator preference. Clear/blue for a clean bright look; yellow when maximum corrosion life is wanted; black for appearance. **Always match the spec.**

HEXAVALENT VS TRIVALENT: THE MOST IMPORTANT DISTINCTION ON THIS STATION

- **Trivalent chromium — Cr(III).** The modern, safer chemistry. **RoHS-compliant** (RoHS restricts hazardous substances in many products). This is what the line is built around. Clear/blue, yellow, and black are all available as trivalent.
- **Hexavalent chromium — Cr(VI).** The legacy chemistry. It builds beautiful, durable chromate films — but **it is a known human carcinogen.** It is *not* RoHS-compliant and is being phased out worldwide.

⚠ **HEADS-UP (SAFETY — THE SINGLE MOST IMPORTANT WARNING IN THIS MANUAL)**

Hexavalent chromium, Cr(VI), is a known carcinogen. The OSHA permissible exposure limit is just **5 µg/m³** of air. Where it is still used at all, it demands a **supplied-air respirator, a full protective suit, and medical surveillance per OSHA 29 CFR 1910.1026.** Its waste is **RCRA hazardous** and must be managed under your facility's permit. **Trivalent Cr(III) is the default for good reason — do not substitute hexavalent for trivalent without full regulatory and safety review and management sign-off.**

☀ **FUN FACT**

That famous "rainbow shimmer" of an old-school yellow chromate? Classic Cr(VI) films were prized for it. Modern trivalent yellows now reproduce a similar look without the carcinogen — one of the quiet success stories of green chemistry in metal finishing.

OPERATOR ESSENTIALS (EXACT NUMBERS BY TYPE)

The single most important control variable here is **pH** — every passivate has a narrow "critical pH window," and outside it the film comes out too thin, too thick, non-adherent, or the wrong color.

PARAMETER	TRI CLEAR/BLUE	TRI YELLOW	TRI BLACK	HEX YELLOW (LEGACY)
Temperature	70–85°F	70–110°F	65–85°F	70–95°F
pH	1.5–2.2	1.8–3.5	2.0–3.5	1.5–2.0
Time	30–60 sec	45–90 sec	30–60 sec	10–30 sec
Cr type	Cr(III) RoHS	Cr(III) RoHS	Cr(III) RoHS	Cr(VI) NOT RoHS
Salt spray (white rust)	72–120 hrs	120–200+ hrs	72–120 hrs	96–240 hrs

★ **REMEMBER**

The rule-card number is **"pH per TDS."** Why not a single number? Because the exact window depends on the specific product. The typical trivalent clear/blue window is **1.5–2.2**, but *your* TDS is the authority. pH is *the* control knob on this station — check it often.

⚙ **TECHNICAL STUFF — WHAT IS "SALT SPRAY"?**

It's an accelerated corrosion test (per ASTM B117): parts sit in a sealed cabinet sprayed with salt fog, and you record how many *hours* until white rust appears. More hours = better protection. It's how the "72–120 hrs" type numbers above are generated and how customers specify corrosion requirements.

💡 **TIP (ACID-ZINC-SPECIFIC)**

Acid zinc's brighter, smoother as-plated deposit often passivates more *uniformly* and looks better than alkaline zinc. **But** acid zinc deposits carry slightly higher internal stress — so **use a passivate formulated for acid zinc, not one designed for alkaline zinc.** They are not interchangeable. Using the wrong one is a common cause of adhesion and appearance problems on this line.

STANDARD OPERATING PROCEDURE (STATION 07)

1. **Confirm the part came clean from the post-plate rinse** (Station 06) with low chloride carry-over. Residual chloride here causes blotchy films and kills bath life.
2. **Confirm any required embrittlement bake was done** if the spec calls for baking before passivation.
3. **Check the passivate bath:** correct type for the spec, **pH in the product's window** (typically 1.5–2.2 for tri clear/blue — verify against TDS), and temperature in range.
4. **Immerse for the specified time** — e.g., 30–60 sec for tri clear/blue; 45–90 sec for tri yellow. Time and pH together control film thickness and color.
5. **Drain over the tank,** then proceed to the final rinse and dry — **Stage 08**, covered on its own page next.

TROUBLESHOOTING (STATION 07)

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Rainbow / iridescent (when it should be clear)	pH too low; immersion time too long	Check pH; reduce time
Pale / thin film	pH too high; bath depleted	Lower pH; replenish per TDS
Blotchy / uneven color	Residual chloride; uneven zinc underneath	Improve post-plate rinse; check plating uniformity
Film rubs off	pH out of range; insufficient cure	Check pH; allow the full 24-hr cure before handling
Passivate bath life short	Chloride drag-in; zinc dissolving into the bath	Improve post-plate rinse quality (Station 06)

💡 **TIP**
Notice how many passivate problems trace back *upstream* to the post-plate rinse and plating uniformity. When this station misbehaves, look backward as well as at the passivate itself.

INTEGRATED SAFETY (STATION 07)

⚠ **HEADS-UP (TRIVALENT)**
Even the "safe" trivalent passivate is **acidic (pH 1.5–3.5) and causes chemical burns.** Wear goggles plus face shield, chemical-resistant gloves, and an apron. Flush skin or eyes with water for **15 minutes** on contact.

⚠ **HEADS-UP (HEXAVALENT — WHERE STILL USED)**
Known carcinogen. OSHA PEL 5 µg/m³. Supplied-air respirator, full protective suit, and medical surveillance per **OSHA 29 CFR 1910.1026.** Waste is **RCRA hazardous** — manage per facility permit. The strong recommendation is to run trivalent.

TEACH-BACK & TOP MISTAKES — STATION 07

TEACH-BACK — CAN YOU ANSWER?

- Why does plated zinc need a passivate — what is "white rust"?
- How is a conversion coating different from electroplating?
- What do the colors tell you about thickness/protection?
- Cr(III) vs Cr(VI) — key safety/regulatory facts for each?
- Why is pH the critical control variable; tri clear/blue window?
- Why use a passivate formulated *for acid zinc*?
- The two ways to wreck a good film during dry & cure?

TOP MISTAKES TO AVOID

- Letting pH drift out of the window (wrong color/thin film).
- Carrying chloride in from a weak post-plate rinse.
- Using an alkaline-zinc passivate on acid zinc.
- Drying above 150°F (cracks trivalent films).
- Handling or testing before the 24-hour cure.
- Confusing trivalent and hexavalent.

SIGN-OFF — STATION 07

I confirm that I can independently: distinguish trivalent from hexavalent chromium and state the hazard and PPE for each; select and verify the correct passivate type and color for the spec; measure and hold pH within the product's critical window; run the immersion, final rinse, dry (≤150°F), and 24-hour cure correctly; diagnose common film defects and trace them upstream when needed; and apply the correct PPE.

Operator: ______________________ Date: ____________ Trainer: ______________________ Date: ____________

STAGE 08 OF 08 · THE FINISH LINE

FINAL RINSE & DRY

THE LAST 60 SECONDS — WHERE A PERFECT PART IS FINISHED, OR QUIETLY RUINED

The chromate film is on. You are one short stage from a finished part. Stage 08 has two simple jobs: **rinse off the leftover passivate chemistry** so it doesn't leave residue or keep reacting, and **dry the part gently** so the soft, fresh conversion film sets without cracking. It feels like the easy part — and that is exactly why parts get wrecked here. Drying too hot or handling too soon undoes everything the line just accomplished.

Straight out of the passivate, the part carries a thin film of acidic chromate solution. The gentle **final rinse** washes that residue away so it can't spot or keep etching the new film; the **gentle dry** then drives off water without overheating. Trivalent conversion films are gel-like as they form and **crack if dried too hot** — cracks open paths for corrosion, defeating the whole point of passivating.

★ **REMEMBER**
The passivate film is **soft when fresh.** Heat cracks it; touching it rubs it off. Patience at Stage 08 protects every step upstream.

OPERATOR ESSENTIALS

STEP	SETTING	WHY IT'S SET THERE
Final rinse	`DI water, ambient, 15–30 sec, gentle immersion`	Removes passivate residue without stripping the fresh film
Dry temperature	`120–150°F max (trivalent)`	**Film cracks above 150°F** — do not over-dry
Cure time	`24 hours minimum before packing/testing`	The soft film needs time to harden fully
Handling	`Clean gloves only`	Bare hands mar the soft film and leave salts/oils

⚠ **HEADS-UP (DON'T RUIN A GOOD FILM AT THE FINISH LINE)**
Two easy ways to destroy a perfect passivate in the last 60 seconds: **drying too hot** (above 150°F cracks trivalent films) and **handling or testing before the 24-hour cure** (the soft film rubs off). Patience here protects everything you did upstream.

STANDARD OPERATING PROCEDURE (STAGE 08)

1. **Final DI rinse:** gentle immersion in ambient DI water for **15–30 seconds.**
2. **Drain** over the rinse tank so you carry off as little water as possible.
3. **Dry gently** in warm air, **120–150°F max** — never hotter, or the trivalent film cracks.
4. **Set aside to cure** — at least **24 hours** before packing, salt-spray testing, or rough handling.
5. **Handle with clean gloves** throughout, and inspect for spotting, haze, or the correct color per spec.

TEACH-BACK CHECKLIST

A trainee is signed off on Stage 08 only when they can, unprompted:

- [ ] State the two jobs of Stage 08 (rinse off passivate residue; gently dry so the film sets).
- [ ] State the maximum dry temperature and what happens above it (**150°F max; the trivalent film cracks above it**).
- [ ] State the final-rinse water, temperature, and time (DI, ambient, 15–30 sec).
- [ ] State the cure time before packing or testing, and why (**24 hours minimum**; the fresh film is soft and needs to harden).
- [ ] Explain why fresh, uncured parts are handled with clean gloves only.

TOP MISTAKES NEW OPERATORS MAKE

1. **Drying too hot to "speed things up."** Above 150°F you crack the film you just grew.
2. **Packing or testing before the 24-hour cure.** The soft film rubs off and salt-spray results read falsely low.
3. **Skipping the final DI rinse,** leaving residue that spots or keeps reacting.
4. **Bare-handed handling** of fresh, soft film — fingerprints and mars.

SIGN-OFF — STAGE 08

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I confirm the part received a final DI rinse, was dried at 150°F or below, was handled with clean gloves, and was set aside for the full 24-hour cure before packing or testing — with required PPE worn."

PART FOUR — BACK MATTER & CERTIFICATION

YOUR DECODER RING

GLOSSARY — EVERY TERM, PLAIN AND SIMPLE

ANYTIME A WORD IN THIS MANUAL MADE YOU PAUSE, FLIP BACK HERE

TIP
Keep this glossary tabbed. The fastest way to sound like a veteran on the shop floor is to use the right word for the right thing.

Acid Activation (acid dip, pickle): Stage 3. A dip in strong acid (HCl or H_2SO_4) that dissolves the invisible oxide on steel and "wakes up" the bare metal so zinc can grip it. *No activation, no adhesion.*

Acid Chloride Zinc: The process this manual is about. "Acid" because the bath is acidic (pH 4.8–5.8); "chloride" because conductivity comes from chloride salts (KCl or NH_4Cl). Plates fast and bright but is fussy about contamination.

Adhesion: How well the plated zinc sticks to the steel. Good adhesion means it won't peel or flake.

Agitation: Movement of the solution or parts. Keeps fresh chemistry against the part, knocks off hydrogen bubbles (which cause pits), and evens out the deposit. Can be air, mechanical, or barrel rotation.

Alkaline: The opposite of acidic; high pH (above 7, often 11–13.5). The soak cleaner is alkaline.

Alkaline Zinc: A *different* high-pH zinc process. Better throwing power and lower embrittlement risk, but slower and lower efficiency (60–80%).

Ambient: Room temperature, roughly 60–85°F. No heating needed.

Amp-Hour (A·h): "How much plating you've done": amps × hours. Brighteners are used up per amp-hour, so you replenish based on amp-hours run, not clock time.

Anode: The positively-charged metal in the tank. In acid zinc we use **dissolving zinc anodes** (99.99% / SHG): as you plate they dissolve and put zinc back into the bath.

Anode Bag: A polypropylene bag around each anode that catches sludge so it doesn't fall in and cause roughness. *Always bag your anodes.*

ASF: Amps per Square Foot — a unit of current density. Acid zinc runs about 10–60 ASF.

ASTM B633: The common industry spec for zinc plating on steel. It sets minimum thickness by environment (see Service Condition) and a separate **Type** that denotes the supplementary finish — **Type I** = as-plated (no chromate), **Type II** = colored (e.g., yellow) chromate, **Type III** = colorless/clear chromate, **Type IV** = phosphate. The Type describes the chromate, *not* whether the bath is acid or alkaline. Our clear-passivated acid zinc is typically specified as **Type III, Service Condition per the print.**

ASTM B850: The spec covering the hydrogen-embrittlement relief bake (post-plate oven treatment for hardened steel).

Barrel Plating: Plating many small parts together by tumbling them in a rotating perforated barrel. Contrast with Rack Plating.

Boric Acid (H_3BO_3): A bath ingredient (20–35 g/L) that acts as a buffer — keeps pH stable right at the part's surface. *Don't neglect it.*

Brightener System: The organic additives that make zinc come out shiny. Acid zinc uses three: Carrier, Primary, and Secondary.

Buffer: A chemical that resists pH change. Boric acid is the buffer in the acid zinc bath.

Burning: A defect: dark, rough, powdery deposit at high-current areas (edges, tips). Caused by pH too low, zinc too low, or too much current.

Carbon Treatment: A bath cleanup where you circulate the solution through activated carbon to pull out organic contamination and broken-down brightener.

Carrier (Conditioner): The first brightener component. Refines grain, prevents pitting, wets the surface. Too little = pitting/haze; too much = haze, dull, foam.

Cathode: The part being plated — the negatively-charged work. Zinc deposits onto the cathode.

Cathode Efficiency: The percentage of current that actually deposits zinc (versus making hydrogen gas). Acid zinc is **95–98%** — its biggest advantage. (Some suppliers cite up to 99%.)

Chelating / Complexing: Chemically "holding onto" metal ions in solution. Chloride salts help complex zinc and assist anode corrosion.

Chloride: The negatively-charged part of the salts (KCl, NH_4Cl) that make the bath conductive. **Corrosive** — it eats mild steel, so equipment must be plastic. Chloride dragged into the passivate ruins it.

Chromate / Conversion Coating: See Passivation. A thin protective film grown chemically (not electroplated) on the zinc surface.

Conductivity (µS/cm): A measure of dissolved salts in rinse water — how "dirty" the rinse is. **Lower is cleaner.** Rinse targets tighten closer to the plating bath.

Counter-Current Rinse: A water-saving setup where fresh water flows opposite to the parts, so the cleanest water meets the cleanest parts.

Cr(III) / Trivalent Chromium: The modern, RoHS-compliant form of chromium used in today's passivates. Much safer than hexavalent.

Cr(VI) / Hexavalent Chromium: The old form. A **known carcinogen** and NOT RoHS-compliant. Legacy use only, under strict OSHA controls.

Current Density (CD): How much current flows per unit of part surface area (see ASF / A/dm²). High-CD areas (edges, tips) and low-CD areas (recesses) behave differently.

DI Water (Deionized Water): Water with dissolved minerals removed. Cleaner than tap; preferred for rinses near the plating and passivate tanks.

Dendrites: Tiny tree-like or whisker-like growths in the deposit — a defect, often from metallic contamination like lead.

Drag-In / Drag-Out: Chemistry that rides along on the parts from one tank to the next. *Drag-out* leaves a tank; *drag-in* arrives in the next. Rinses exist to control it.

Dummy / Dummying: A cleanup trick: plate onto scrap "dummy" cathodes at low current to draw metallic contamination (Cu, Pb) out of the bath.

Filtration: Continuously pumping the bath through a filter to remove particles. Acid zinc runs continuous filtration, 2–4 turnovers/hr, often with a carbon canister.

• GLOSSARY (CONTINUED)

Hexavalent: See Cr(VI).
Hull Cell: A small, trapezoid-shaped test tank that plates a sample panel across a *range* of current densities at once. Reading the panel tells you what's wrong with the bath. Standard test: **267 mL, 2 amps, 10 minutes.**
Hydrogen Embrittlement (HE): When atomic hydrogen soaks into steel during pickling/plating and makes hardened steel **brittle** — it can crack later under load, sometimes catastrophically. The acid pickle is the single greatest source on this line.
HE Relief Bake: Baking plated parts to drive hydrogen back out before it cracks. For steel **>40 HRC**: bake **within 4 hours** of plating at **375 ±25°F for at least 8 hours** (per ASTM B850).
HRC (Rockwell C Hardness): A hardness scale for steel. Above **40 HRC**, steel is "high-strength" and at real risk of hydrogen embrittlement — needs special care.
Inhibitor: An additive in the acid pickle that slows attack on the base metal and reduces hydrogen absorption. **Required for >40 HRC steel.**
Iron (Fe) Contamination: Dissolved iron that drags into the plating bath from the pickle. Causes haze and dark deposits. **Action level ~10 ppm.** Unlike alkaline zinc, acid zinc can't precipitate and filter iron out — it accumulates, so the pre-plate rinse is your main defense.
KCl (Potassium Chloride): The preferred conductive salt for modern acid zinc. **No ammonia** in the wastewater. Costs a bit more than NH_4Cl but is environmentally favored. Typical 120–250 g/L.
Leveling: The bath's ability to fill in tiny scratches and produce a smoother surface than the bare steel. Acid zinc levels very well.
LOTO (Lockout/Tagout): De-energizing and locking the power off before working on the rectifier or electrical equipment, so it can't be switched on while someone's hands are in it.
µm (Micrometer / Micron): A unit of coating thickness. One micron = 0.001 mm ≈ 0.04 mil. Acid zinc deposits typically run **5–15 µm.**
Mil: Another thickness unit: one-thousandth of an inch (0.001″). 1 mil ≈ 25 µm.
NH_4Cl (Ammonium Chloride): An older conductive salt for acid zinc. Slightly higher efficiency and lower cost, but produces **ammonia in the wastewater** (a regulatory headache). Being phased out in favor of KCl.
Oxide: The thin film (rust/tarnish) that forms on bare steel in air. It blocks adhesion, so acid activation must dissolve it right before plating.
Passivation (Passivate / Chromate): Stage 7, the final chemical station. A trivalent-chromium conversion coating grown on the bare zinc that dramatically slows **white rust** and can add color. *Zinc without passivate is a coating waiting to white rust.*
pH: A 0–14 scale of acidity. Below 7 = acidic. The plating bath sits at **4.8–5.8**; trivalent clear/blue passivate at roughly **1.5–2.2.**
Pitting: Tiny holes or craters in the deposit, usually from hydrogen bubbles clinging to the part (low carrier or poor agitation) or from dirty parts.
ppm (Parts Per Million): A unit for very small concentrations, used for contamination limits (e.g., copper action level **0.5 ppm**).
Primary Brightener: The second brightener component. Delivers overall brightness across the current-density range. Too little = dull (especially mid-CD); too much = brittle, peeling, high stress.
Rack Plating: Plating parts individually hung on a rack/fixture (contrast with Barrel). Used for larger or cosmetic parts.
Rectifier: The big power supply that converts AC wall power into the DC current that drives plating. **High amperage — shock and arc-flash hazard.** Always LOTO before servicing.
RoHS: "Restriction of Hazardous Substances," a regulation that bans certain hazardous materials (including hexavalent chromium) in many products. Trivalent passivates are RoHS-compliant; hexavalent is not.
Salt Spray Test: A corrosion test that sprays parts with salt fog and measures the hours until rust appears. Used to verify passivate performance (e.g., trivalent clear/blue: 72–120 hours to white rust).
Secondary Brightener: The third brightener component. Provides brightness and leveling specifically at **high current density** (edges, tips). Too little = dull at high CD; too much = burning at low CD and brittle edges.
Service Condition (SC): ASTM B633's rating of how harsh the part's environment is, setting the minimum zinc thickness: SC 1 (indoor mild, 5 µm) · SC 2 (indoor moderate, 8 µm) · SC 3 (outdoor moderate, 12 µm) · SC 4 (severe/marine, 25 µm).
SHG (Special High Grade) Zinc: 99.99%-pure zinc anodes. Purity matters because impurities like lead in the anode contaminate the bath.
Skimmer: A device that removes floating oil off the top of the soak-clean tank so it doesn't redeposit on parts.
Smut: A dark, loose film left on steel after pickling, usually from high dissolved iron or copper in the acid. Fix: dump/recharge the acid.
Throwing Power: The bath's ability to plate evenly into recesses. Acid zinc's throwing power is **poor** — it covers recesses worse than alkaline zinc, and this is acid zinc's main weakness. The shop figures (surface-to-recess thickness ratio, about 3:1 to 5:1 for acid vs. 1.5:1 to 2:1 for alkaline) read "better" the *closer they are to 1:1*, so alkaline's smaller spread is the more even one.
Titrate / Titration: A chemical test that measures the concentration of a solution (e.g., cleaner strength) so you know whether to add more. Titrate the cleaner at least weekly.
Trivalent: See Cr(III).
Turnover (Tank Turnover): One complete pass of the entire tank volume through the filter (or one full replacement of rinse volume). "4–6 turnovers/hour" means the whole tank cycles 4–6 times an hour.
Water Break Test: The free, reliable test for clean steel: dip and pull the part, watch the water. If it **sheets** off in an unbroken film, the part is clean. If it **beads up**, there's still oil — reclean.
White Rust: The white, powdery corrosion of bare zinc (zinc oxide/hydroxide). The whole point of passivation is to delay it.
Zinc (as $ZnCl_2$ / Zinc Metal): The metal you're depositing. Bath concentration is given as zinc metal: **15–45 g/L (2–6 oz/gal)**, replenished by the dissolving anodes.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

★ **REMEMBER**
Most acid zinc plating defects trace back to **three root causes**: a contaminated bath (metals or organics), a brightener out of balance, or a rinse that let drag-in through. Diagnose with a **Hull Cell** before you start dumping chemistry.

⚠ **HEADS-UP**
Before changing acid strength, pickle time, or anything that increases metal attack, check whether the part is **>40 HRC** — those changes raise hydrogen-embrittlement risk. When in doubt, ask your supervisor and lean on the relief bake.

STATION 01 — SOAK CLEAN

SYMPTOM	LIKELY CAUSE	FIX
Water break test fails (water beads)	Low cleaner concentration or low temp	Titrate cleaner; increase temperature
Oily film remains after cleaning	Floating oil redepositing on parts	Clean/install surface skimmer; dump bath if heavily contaminated
White residue on parts	Hard-water salts or cleaner residue	Check rinse-water quality (use DI)
Etching or discoloration of the part	Cleaner too aggressive or temp too high	Reduce temperature; switch to milder formulation
Spotty adhesion shows up later	Uneven cleaning; air pockets on rack	Reposition parts on rack; add agitation

STATION 02 — RINSE (PRE-ACTIVATION)

SYMPTOM	LIKELY CAUSE	FIX
Rinse conductivity high (>500 μS/cm)	Too much drag-out; low flow	Increase drain time off the cleaner; increase rinse flow
Soapy residue on parts	Inadequate rinse time	Extend dwell time; add agitation
Acid activation tank goes alkaline	Cleaner carrying over (drag-in)	Improve rinse; add a second rinse stage

STATION 03 — ACID ACTIVATION (PICKLE)

SYMPTOM	LIKELY CAUSE	FIX
Dark oxide still on surface	Acid too dilute; dip too short	Increase concentration; extend time **(check HE risk first)**
Pitting / etching of base metal	Acid too concentrated, too long, or too warm	Reduce concentration; shorten dip; add inhibitor
Adhesion fails downstream	Incomplete activation / leftover oxide	Re-check acid strength and time; check for alkaline drag-in
Excessive HCl fuming	High concentration/temp; poor ventilation	Reduce to minimum effective conc.; check exhaust
Dark smut on surface	High dissolved iron; copper contamination	Dump and recharge acid; find the copper source
HE cracking/failures	Over-long pickle; no inhibitor; >40 HRC steel	Shorten pickle; add inhibitor; review hardness; consider alkaline zinc

STATION 04 — RINSE (PRE-PLATE / GUARD RINSE)

SYMPTOM	LIKELY CAUSE	FIX
Rinse conductivity high (>200 μS/cm)	Heavy acid drag-out; low flow	Increase drain time off pickle; increase flow
Plating bath pH dropping fast	Acid drag-in from pickle	Tighten rinse conductivity; add double rinse
Iron climbing in plating bath	Dissolved Fe dragging in from spent pickle	Dump/refresh pickle; improve rinse; add second stage
Parts oxidizing before they plate	Transfer too slow; parts drying out	Speed up — into the plating bath within 30 sec

STATION 05 — ACID ZINC PLATING

SYMPTOM	LIKELY CAUSE	FIX
Burning at high CD (dark, rough edges/tips)	pH too low; zinc too low; excess secondary brightener	Raise pH; add zinc; run a Hull Cell
Dull deposit overall	Primary brightener depleted; temp too low; organics	Add primary per amp-hour; check temp; carbon treat
Pitting across the part	Low carrier; poor agitation; dirty parts	Add carrier; increase air agitation; check cleaning
Hazy deposit	Iron contamination; organics; excess carrier	Analyze Fe; carbon treat; check carrier level; dummy
Dark streaks or spots	Copper or lead contamination; anode sludge	Analyze metals; dummy plate; check anode bags
Roughness	Particulate; lead; poor filtration	Check/replace filter; analyze Pb; bag anodes
Peeling / poor adhesion	Excessive internal stress; brightener imbalance	Hull Cell bend test; reduce primary; analyze Cu
Poor coverage in recesses	Normal for acid zinc (throwing power 3:1–5:1); pH too low	Optimize pH; accept some variation vs alkaline

STATION 06 — RINSE (POST-PLATE)

SYMPTOM	LIKELY CAUSE	FIX
Passivate bath life short	Chloride dragging in from plating bath	Tighten rinse; add a second stage
Blotchy passivate	Residual chloride still on the zinc	Improve rinse agitation and time
White haze on the zinc	Zinc oxidizing; hard water	Use DI water; reduce transfer time

STATION 07 — PASSIVATE (TRIVALENT CHROMATE)

SYMPTOM	LIKELY CAUSE	FIX
Rainbow/iridescent when it should be clear	pH too low; time too long	Check pH; reduce time
Pale or thin film	pH too high; bath depleted	Lower pH; replenish per TDS
Blotchy / uneven color	Residual chloride; uneven zinc deposit	Improve post-plate rinse; check plating uniformity
Film rubs off	pH out of range; not enough cure time	Check pH; allow 24-hour cure before handling/testing
Passivate bath life short	Chloride drag-in; zinc dissolving in bath	Improve post-plate rinse

HULL CELL QUICK-READ (267 ML · 2 A · 10 MIN)

PANEL PATTERN	LIKELY CAUSE
Bright across the whole panel	Bath is in balance
Dull mid-range, bright high CD	Low **primary** brightener
Dull at high CD only	Low **secondary** brightener
Hazy overall	Organic or metallic contamination
Dark/burned at high CD	pH too low; zinc too low
Pitting across the panel	Low carrier; poor filtration

> 💡 **TIP**
> Notice how many passivate problems trace back *upstream* to the post-plate rinse and plating uniformity. Most downstream defects are born upstream. Fix the root, not the symptom.

TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

★ **REMEMBER**

An operator is "line-qualified" only when every station they run is signed. Nobody touches the **rectifier**, the **acid pickle**, or the **passivate** solo until the **Safety / PPE & SDS** column is signed. Safety qualification comes first, always.

Sign-off levels: O = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	Safety / PPE & SDS (all stations)	____	__________	__________	______
2	Station 01 — Soak Clean + water break test	____	__________	__________	______
3	Station 02 — Rinse (Pre-Act) + conductivity	____	__________	__________	______
4	Station 03 — Acid Activate (incl. HE check)	____	__________	__________	______
5	Station 04 — Rinse (Pre-Plate / Guard)	____	__________	__________	______
6	Station 05 — Acid Zinc Plating operation	____	__________	__________	______
7	Hull Cell / bath test — run & interpret	____	__________	__________	______
8	Station 06 — Rinse (Post-Plate)	____	__________	__________	______
9	HE Bake (ASTM B850) decision & logging	____	__________	__________	______
10	Station 07 — Passivate (trivalent chromate)	____	__________	__________	______
11	Stage 08 — Final Rinse / Dry & cure	____	__________	__________	______
12	LOTO on rectifier	____	__________	__________	______
13	Emergency response — eyewash, shower, spill	____	__________	__________	______
14	Waste / drag-out handling	____	__________	__________	______
15	Troubleshooting — symptom recognition	____	__________	__________	______

Operator name: ______________________________ **Hire/start date:** __________________

Supervisor name: ______________________________ **Review cycle:** ☐ Annual ☐ Other: ______________

⚠ **HEADS-UP**

Safety rows (1, 12, 13) are not optional and not "fill in later." An operator may not work *any* station until PPE/SDS, LOTO, and emergency response are signed off. Safety competency comes before production competency, always.

💡 **TIP**

Re-verify each operator annually, and **immediately** after any process or chemistry change. Note the recertification date in the cell (e.g., "Q 6/26 · recert 6/27"). People drift from procedure over time, and a yearly refresher catches bad habits before they become defects — or injuries.

20 QUESTIONS · PASSING SCORE 80% (16/20)

ACID ZINC COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

★ **REMEMBER**

Passing score: 80% (16 of 20 correct). Any **safety** question answered wrong (Q5, Q6, Q14, Q19) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: ______________________ **Date:** ______________ **Score:** ________ / 20 **Safety-gate Qs (all must be correct): 5, 6, 14, 19**

Q1. Acid chloride zinc gets its high conductivity from chloride salts. Which two salts are used?
A. Sodium hydroxide and boric acid B. Potassium chloride or ammonium chloride C. Sulfuric acid or nitric acid D. Sodium cyanide or zinc cyanide

Q2. (Short answer) What is the **water break test**, and what result tells you a part is clean?

Q3. The soak-clean tank typically runs at:
A. 70–85°F B. 130–160°F C. 200–250°F D. Below freezing

Q4. (Short answer) Name the main job of the **acid activation (pickle)** stage. In one sentence, why does the manual say "no activation, no adhesion"?

Q5. [SAFETY GATE] Which station is described as the **single greatest source of hydrogen embrittlement** on the acid zinc line?
A. Soak clean B. Acid activation (pickle) C. Post-plate rinse D. Passivate

Q6. [SAFETY GATE] A part is **>40 HRC** (high-strength steel). Per ASTM B850, the hydrogen-embrittlement relief bake is:
A. 200°F for 1 hour, any time within a week B. 375 ±25°F for at least 8 hours, started within 4 hours of plating C. 500°F for 30 minutes D. No bake is needed for zinc plating

Q7. (Short answer) The pre-activation rinse (Station 02) targets **< 500 µS/cm**, but the pre-plate rinse (Station 04) targets **< 200 µS/cm**. Why are the rinses closer to the plating bath held to a *tighter* standard?

Q8. The acid zinc **plating bath** operates at what pH?
A. 1.5–2.2 B. 4.8–5.8 C. 7.0 (neutral) D. 11.0–13.5

Q9. (Short answer) What is the job of **boric acid** in the plating bath, and why shouldn't you neglect it?

Q10. Acid zinc's cathode efficiency is about:
A. 40–50% B. 60–80% C. 95–98% D. 100%, always

Q11. (Short answer) Name the **three components** of the acid zinc brightener system and, in a few words, what each one does.

Q12. You run a Hull Cell and the panel is **dull at high current density only**. The most likely cause is:
A. Low primary brightener B. Low secondary brightener C. Iron contamination D. pH too high

Q13. Which metallic contaminant has the **lowest action level** (most sensitive) in the acid zinc bath?
A. Iron (Fe) at 10 ppm B. Copper (Cu) at 0.5 ppm C. Zinc at 45 g/L D. Boric acid

Q14. (Short answer) **[SAFETY GATE]** Why must acid zinc tanks, heaters, and bus bars be plastic (PP/PVC) rather than mild steel?

Q15. Why must the **post-plate rinse** be especially good before the passivate?
A. To warm the parts up B. To wash off chloride drag-out, which poisons the passivate bath C. To add brightener D. It isn't important

Q16. The final station, **passivation**, uses which type of chromium in modern RoHS-compliant lines?
A. Hexavalent — Cr(VI) B. Trivalent — Cr(III) C. Cyanide D. No chromium at all

Q17. (Short answer) What does passivation actually *do* for the zinc, and what corrosion is it protecting against?

Q18. A trivalent clear/blue passivate's critical pH window is roughly:
A. 1.5–2.2 B. 4.8–5.8 C. 7.0 D. 11–13

> ⚠ **SAFETY SECTION — MUST BE CORRECT TO CERTIFY**
> Questions 5, 6, 14, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

Q19. (Short answer / Safety) **[SAFETY GATE]** List **four** pieces of PPE you must wear when working at the **acid activation** station, and name the procedure you must perform before doing any maintenance on the **rectifier**.

Q20. (Short answer) Acid zinc plates faster and brighter than alkaline zinc — but name **two** trade-offs (weaknesses) of acid zinc compared to alkaline zinc.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

> ⚠ **HEADS-UP**
> Keep this page out of the trainee's copy. Questions **5, 6, 14, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

Q1. B — Potassium chloride (KCl) or ammonium chloride (NH_4Cl).

Q2. Dip and pull the part and watch the water. If it **sheets off in an unbroken, continuous film**, the part is clean. If it **beads up**, oil remains — reclean. (It's free; use it.)

Q3. B — 130–160°F (54–71°C); do not exceed 180°F.

Q4. It **dissolves the surface oxide/rust** and produces a chemically active steel surface so zinc can bond. Without it, oxides block the bond and the plating won't stick (no adhesion).

Q5. B — Acid activation (pickle).

Q6. B — 375 ±25°F for at least 8 hours, started within 4 hours of plating (ASTM B850).

Q7. Because tighter rinsing closer to the plating bath limits **drag-in** of acid and dissolved **iron** from the pickle. Acid drag-in depresses bath pH and breaks down brighteners, and iron can't be filtered out of an acid zinc bath — it accumulates. The closer to the plating tank, the more damage drag-in does, so the standard tightens. (Full credit for naming pH protection, brightener protection, *or* iron control.)

Q8. B — 4.8–5.8.

Q9. Boric acid is the **pH buffer** — it keeps the pH stable right at the part's surface (the cathode film) during plating, where the chemistry tries to drift. Neglect it and you get inconsistent deposits, burning, and brightness problems.

Q10. C — 95–98% (suppliers may cite up to 99%).

Q11. Carrier (conditioner) — grain refinement, anti-pit, wetting; **Primary brightener** — overall brightness across the CD range; **Secondary brightener** — high-CD brightness and leveling (edges/tips).

Q12. B — Low secondary brightener.

Q13. B — Copper (Cu) at 0.5 ppm (lead is also 0.5 ppm; iron is far higher at 10 ppm). Accept "Cu."

Q14. Because **chloride is corrosive** and attacks/eats mild steel. Steel tanks and equipment would corrode, contaminating the bath with iron and failing structurally — so PP/PVC plastic is used.

Q15. B — To wash off chloride drag-out; chloride dragged into the passivate poisons/degrades the bath and causes thin, blotchy coatings.

Q16. B — Trivalent, Cr(III). (Hexavalent, Cr(VI), is a known carcinogen and not RoHS-compliant.)

Q17. It grows a **chromate conversion coating** on the bare zinc that dramatically slows corrosion — specifically **white rust** (the white powdery corrosion of zinc) — and can add color. "Zinc without passivate is a coating waiting to white rust."

Q18. A — 1.5–2.2.

Q19. PPE (any four): sealed chemical splash goggles (*not* safety glasses — do not accept "safety glasses" on this acid station), face shield, acid-resistant gloves, chemical apron, chemical-resistant boots, respirator if ventilation is inadequate. **Rectifier procedure: LOTO (Lockout/Tagout)** — de-energize and lock out the power before any rectifier maintenance.

Q20. Any two of: **poor throwing power** (covers recesses worse than alkaline — surface-to-recess ratio ~3:1–5:1 vs alkaline's more-even 1.5:1–2:1; closer to 1:1 is better); **more sensitive to metallic contamination** (Fe, Cu, Pb); **higher hydrogen-embrittlement risk** per unit zinc; **chloride is corrosive to equipment**; **ammonium-chloride baths add ammonia to wastewater**; **more complex/sensitive brightener chemistry**.

> ★ **REMEMBER**
> A score of 16/20 passes — but every safety question (5, 6, 14, 19) must be correct to certify. If a trainee misses a safety item, re-teach it and re-test before sign-off. We don't gamble with people.

PRINT, FILL IN, SIGN

PLATING POSTERS

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CERTIFICATE OF COMPLETION

ACID ZINC PLATING TRAINING PROGRAM

This certifies that

____

OPERATOR NAME

has successfully completed the **Acid Zinc Plating** training course — covering all eight process stages (Clean, Rinse, Activate, Rinse, Plate, Rinse, Passivate, and Final Rinse/Dry), bath chemistry and maintenance, Hull Cell diagnostics, hydrogen-embrittlement control, troubleshooting, and the safety practices for alkaline cleaners, mineral acids, the acid zinc bath, and chromate passivation — and has passed the Completion Test with a score of ________ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: ________________ **Re-certification due:** ________________

____

TRAINEE SIGNATURE

____

CERTIFYING TRAINER

____

SUPERVISOR